Introduction to Analysis II

Lecture notes, Chapters 6–10

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Chapter 6. Sequences and Series of Functions

6.1 Uniform convergence

Motivation. Pointwise limits can destroy continuity (x^n on $[0,1]$); uniform limits preserve it.

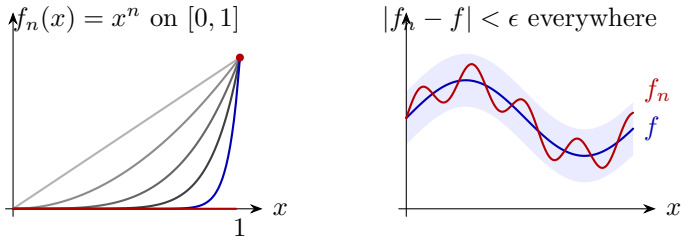


Figure. *Left: x^n has a discontinuous pointwise limit on $[0,1]$, so convergence is not uniform. Right: uniformly, the whole graph of f_n eventually lies in the ϵ-band around f.*

Let $X \subset \mathbb{R}^n$ and $f_n, f : X \to \mathbb{R}^m$ (or $\mathbb{C}$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each x, and **uniformly** if $\|f_n - f\|_{\sup} = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$.

Example 6.1

$x^n \to 0$ pointwise on $[0,1)$ but not uniformly, since $\sup_{[0,1)} x^n = 1$; the limit of x^n on $[0,1]$ is discontinuous. $f_n(x) = nx(1-x^2)^n$ tends to 0 pointwise on $[0,1]$, while $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

Theorem 6.1 (6.1.1)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Given x_0 and ϵ, choose N with $\|f_N - f\|_{\sup} < \epsilon/3$ and δ with $\|f_N(x) - f_N(x_0)\| < \epsilon/3$ for $\|x - x_0\| < \delta$. Then

$$\|f(x) - f(x_0)\| \le \|f(x) - f_N(x)\| + \|f_N(x) - f_N(x_0)\| + \|f_N(x_0) - f(x_0)\| < \epsilon.$$

■

Theorem 6.2 (6.1.2, uniform Cauchy criterion)

(f_n) converges uniformly on X iff for every $\epsilon > 0$ there is N with $\|f_m(x) - f_n(x)\| < \epsilon$ for all $m, n \ge N$ and $x \in X$.

Proof. Necessity is the triangle inequality. Conversely, each $(f_n(x))$ is Cauchy, so converges to some $f(x)$ (Theorem 2.14 of Volume I); letting $m \to \infty$ in $\|f_m(x) - f_n(x)\| < \epsilon$ gives $\|f(x) - f_n(x)\| \le \epsilon$ for all x and $n \ge N$. ■

Corollary 6.3 (6.1.3, Weierstrass M-test)

If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on X.

Proof. For $n > m$, $|\sum_{k=m+1}^{n} f_k(x)| \le \sum_{k=m+1}^{n} M_k$, which is small uniformly in x; apply Theorem 6.2 to the partial sums. ■

Example 6.2

$\sum 2^{-n}\cos(3^n x)$ converges uniformly, so it is continuous. Weierstrass's function $\sum a^n \cos(b^n \pi x)$ ($0 < a < 1$, $ab > 1 + 3\pi/2$, b odd) is continuous but nowhere differentiable. Also, there is a continuous curve filling a square (Peano), obtained as a uniform limit.

6.2 Differentiation and integration of limits

Motivation. Limits pass under integrals uniformly; under derivatives, the derivatives must converge uniformly.

Theorem 6.4 (6.2.1)

Let $f_n : [a, b] \to \mathbb{R}$ satisfy:
(a) each f_n is differentiable;
(b) $(f_n(x_0))$ converges for some $x_0 \in [a, b]$;
(c) (f_n') converges uniformly. Then (f_n) converges uniformly to a differentiable f, and $f'(x) = \lim f_n'(x)$.

Proof. By the mean value theorem applied to $f_m - f_n$,

$$|(f_m - f_n)(x) - (f_m - f_n)(t)| \le |x - t| \, \|f_m' - f_n'\|_{\sup}. \qquad (*)$$

With $t = x_0$: (f_n) is uniformly Cauchy, so $f_n \to f$ uniformly. Fix x; for $t \neq x$ let $\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}$, $\phi(t) = \frac{f(t) - f(x)}{t - x}$. By $(*)$, $\phi_n \to \phi$ uniformly on $[a, b] - \{x\}$. With $g = \lim f_n'$, choose N with $|\phi_N - \phi| < \epsilon$ and $|f_N'(x) - g(x)| < \epsilon$, then δ with $|\phi_N(t) - f_N'(x)| < \epsilon$ for $0 < |t - x| < \delta$. Then $|\phi(t) - g(x)| < 3\epsilon$, so $f'(x) = g(x)$. ■

Uniform convergence of (f_n) alone is not enough: $\sin(nx)/\sqrt{n} \to 0$ uniformly, but the derivatives $\sqrt{n}\cos(nx)$ diverge.

Proposition 6.5 (6.2.2)

If Riemann integrable $f_n : [a, b] \to \mathbb{R}$ converge uniformly to f, then f is integrable and $\int_a^b f = \lim \int_a^b f_n$.

Proof. Let $\epsilon_n = \|f_n - f\|_{\sup}$. From $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, the upper and lower integrals of f lie within $\epsilon_n(b - a)$ of $\int f_n$; so they differ by at most $2\epsilon_n(b - a) \to 0$, and $|\int f - \int f_n| \le \epsilon_n(b - a)$. ■

Example 6.3

Integrating $\frac{1}{1+t^2} = \sum_{n \ge 0}(-1)^n t^{2n}$ (uniform on $[-r, r]$, $r < 1$) gives $\arctan x = \sum (-1)^n x^{2n+1}/(2n + 1)$ for $|x| < 1$; similarly $\log(1 + x) = \sum (-1)^{n+1} x^n/n$. If $a_n > 0$ decrease and $\sum a_n < \infty$, then $na_n \to 0$: $\frac{n}{2} a_n \le a_{\lceil n/2 \rceil} + \cdots + a_n \to 0$.

6.3 Double sequences and double series

Motivation. Conditions, above all absolute convergence, under which iterated limits and sums commute.

Proposition 6.6 (6.3.1)

Let (a_{mn}) be a double sequence such that (a) $a_{mn} \to b_m$ as $n \to \infty$ for each m;
(b) $a_{mn} \to c_n$ as $m \to \infty$ for each n;
(c) the convergence in (b) is uniform in n: for every ϵ there is N with $|a_{mn} - c_n| < \epsilon$ for $m \geq N$ and all n. Then (b_m) and (c_n) converge to the same limit: $\lim_m \lim_n a_{mn} = \lim_n \lim_m a_{mn}$.

Proof. On $X = \{1/n\} \cup \{0\}$ let $f_m(1/n) = a_{mn}$, $f_m(0) = b_m$; f_m is continuous by (a), and uniformly convergent on X by (c) and Theorem 6.2 (at 0 since $|b_m - b_k| \leq \sup_n |a_{mn} - a_{kn}|$). The limit is continuous (Theorem 6.1) with values c_n at $1/n$ and $\lim b_m$ at 0, so $\lim_n c_n = \lim_m b_m$. ■

Proposition 6.7 (6.3.2)

If continuous $f_n : X \to Y$ converge uniformly to f and $x_n \to x$ in X, then $f_n(x_n) \to f(x)$.

Proof. $\|f_n(x_n) - f(x)\| \leq \|f_n - f\|_{\sup} + \|f(x_n) - f(x)\|$, and f is continuous (Theorem 6.1). ■

With pointwise convergence this fails: $f_n(x) = x^n$, $x_n = 1 - 1/n$, gives $f_n(x_n) \to 1/e$ while $f(1) = 1$.

Proposition 6.8 (6.3.3)

If $\sum_n |a_{mn}| = b_m < \infty$ for each m and $\sum_m b_m < \infty$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$, all series converging absolutely.

Proof. Each $\sum_m |a_{mn}| \leq \sum_m b_m$, so the inner sums converge absolutely.

Let $S_{MN} = \sum_{m \leq M} \sum_{n \leq N} a_{mn}$. As $N \to \infty$, $S_{MN} \to \sum_{m \leq M} \sum_n a_{mn}$; as $M \to \infty$, $S_{MN} \to \sum_{n \leq N} \sum_m a_{mn}$, uniformly in N, since

$$\left| S_{MN} - \sum_{n \leq N} \sum_m a_{mn} \right| \leq \sum_{m > M} b_m.$$

Proposition 6.6 gives the equality of the iterated limits. ■

Proposition 6.9 (6.3.4)

If $a_{mn} \geq 0$ for all m, n, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ in $[0, \infty]$.

Proof. For all M, N, $\sum_{m \leq M} \sum_{n \leq N} a_{mn} \leq \sum_{n \leq N} \sum_m a_{mn} \leq \sum_n \sum_m a_{mn}$. Letting $N \to \infty$ and then $M \to \infty$ gives $\leq$; the reverse is symmetric. ■

Both sides also equal $\sup_F \sum_{(m,n) \in F} a_{mn}$ over finite sets F, and any enumeration of the double series of nonnegative terms has this sum (Proposition 2.16 of Volume I). For $(a_k)_{k \in \mathbb{Z}}$, $\sum_{k=-\infty}^{\infty} a_k$ means $\lim_{m,n \to \infty} \sum_{k=-m}^{n} a_k$ when this double limit exists.

Corollary 6.10 (6.3.5)

If $a_k \geq 0$ for all $k \in \mathbb{Z}$, then $\lim_n \sum_{-n}^{n} a_k = \lim_n \lim_m \sum_{-m}^{n} a_k = \lim_m \lim_n \sum_{-m}^{n} a_k$ in $[0, \infty]$.

Proof. All three equal $\sup_{m,n} \sum_{-m}^{n} a_k$, since the partial sums increase in both m and n. ■

Corollary 6.11 (6.3.6)

If $\sum_{k \in \mathbb{Z}} |a_k| < \infty$, the three limits in Corollary 6.10 exist and coincide.

Proof. Apply Corollary 6.10 to $a_k^{\pm} = \max(\pm a_k, 0)$, whose sums are finite, and subtract. ∎

6.4 Power series

Motivation. Power series converge uniformly inside their radius and differentiate termwise, defining e^x, $\sin x$, $\cos x$.

Theorem 6.12 (6.4.1, Cauchy–Hadamard)

Let $\lambda = \limsup|a_n|^{1/n}$ and $R = 1/\lambda$ ($R = \infty$ if $\lambda = 0$, $R = 0$ if $\lambda = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$; for every $0 < r < R$ it converges uniformly on $[-r, r]$. Its sum f is differentiable on $(-R, R)$ with $f'(x) = \sum n a_n x^{n-1}$, and hence C^∞, with $a_n = f^{(n)}(0)/n!$.

Proof. $\limsup|a_n x^n|^{1/n} = \lambda|x|$; apply the root test (Proposition 2.19 of Volume I). On $[-r, r]$ use the M-test with $M_n = |a_n| r^n$, summable as $r < R$. Since $\limsup|n a_n|^{1/n} = \lambda$ ($n^{1/n} \to 1$), the derived series has the same radius and converges uniformly on $[-r, r]$; Theorem 6.4 gives $f' = \sum n a_n x^{n-1}$. Iterate. ∎

Theorem 6.13 (6.4.2, re-expansion)

If $f(x) = \sum a_n x^n$ on $(-R, R)$ and $|c| < R$, then $f(x) = \sum_n \frac{f^{(n)}(c)}{n!}(x-c)^n$ for $|x-c| < R - |c|$.

Proof. $f(x) = \sum_n a_n (c + (x-c))^n = \sum_n \sum_{k \le n} a_n \binom{n}{k} c^{n-k}(x-c)^k$. Since $\sum_n \sum_k |a_n| \binom{n}{k} |c|^{n-k}|x-c|^k = \sum_n |a_n|(|c| + |x-c|)^n < \infty$, Proposition 6.8 allows the order of summation to be exchanged: $f(x) = \sum_k (\sum_n \binom{n}{k} a_n c^{n-k})(x-c)^k$, and the coefficient of $(x-c)^k$ is $f^{(k)}(c)/k!$ by termwise differentiation (Theorem 6.12). ∎

So a power series is **analytic** (locally a power series) throughout its interval of convergence. The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is $\sum c_n$, $c_n = \sum_{k=0}^{n} a_k b_{n-k}$.

Proposition 6.14 (6.4.3, Mertens)

If $\sum a_n = A$ converges absolutely and $\sum b_n = B$, then $\sum c_n = AB$.

Proof. With partial sums A_n, B_n, C_n and $\beta_n = B_n - B$, $C_n = \sum_{k \le n} a_k B_{n-k} = A_n B + \gamma_n$, $\gamma_n = \sum_{k \le n} a_k \beta_{n-k}$. With $\alpha = \sum |a_k|$ and $|\beta_m| < \epsilon$ for $m \ge N$, $|\gamma_n| \le \epsilon\alpha + \sum_{n-N < k \le n} |a_k||\beta_{n-k}|$, and the last sum $\to 0$ ($a_k \to 0$, β bounded). So $C_n \to AB$. ∎

Theorem 6.15 (6.4.4, Abel)

Let $f(x) = \sum a_n x^n$ on $(-R, R)$. If $\sum a_n R^n$ converges, then $\lim_{x \to R^-} f(x) = \sum a_n R^n$.

Proof. Take $R = 1$ ($a_n \to a_n R^n$) and $\sum a_n = 0$ (subtract a constant). With partial sums $s_n \to 0$, summation by parts gives $\sum_{n \le N} a_n x^n = (1-x)\sum_{n<N} s_n x^n + s_N x^N$, so $f(x) = (1-x)\sum s_n x^n$. If $|s_n| < \epsilon$ for $n \ge M$, then $|f(x)| \le (1-x)\sum_{n<M}|s_n| + \epsilon < 2\epsilon$ for x near 1^-. ∎

Corollary 6.16 (6.4.5)

If $\sum a_n = A$, $\sum b_n = B$ and their Cauchy product $\sum c_n = C$ all converge, then $C = AB$.

Proof. For $|x|<1$ the series $\sum a_n x^n$, $\sum b_n x^n$ converge absolutely, so $\sum c_n x^n = (\sum a_n x^n)(\sum b_n x^n)$ by Proposition 6.14. Let $x \to 1^-$ and use Theorem 6.15 three times. ∎

Example 6.4

Abel's theorem gives $\log 2 = 1 - \tfrac12 + \tfrac13 - \cdots$ and $\tfrac{\pi}{4} = 1 - \tfrac13 + \tfrac15 - \cdots$ from Example 6.3. Defining $e^x = \sum x^n/n!$, Proposition 6.14 gives $e^{x+y} = e^x e^y$; with $\sin, \cos$ defined by their series, $(\sin)' = \cos$, $(\cos)' = -\sin$, and $\sin^2 + \cos^2 = 1$ follow by termwise differentiation.

6.5 Trigonometric series

Motivation. Uniform convergence of trigonometric series, preparing Chapter 9.

A complex vector space V with a map $\langle \cdot, \cdot \rangle$ that is linear in the first variable, satisfies $\langle y, x\rangle = \overline{\langle x, y\rangle}$, and $\langle x, x\rangle > 0$ for $x \neq 0$, is an **inner product space**; $\|x\| = \langle x, x\rangle^{1/2}$.

Theorem 6.17 (6.5.1, Cauchy–Schwarz)

$|\langle x, y\rangle| \le \|x\|\|y\|$ in a complex inner product space; hence $\|x+y\| \le \|x\| + \|y\|$.

Proof. If $y \neq 0$, put $\lambda = \langle x, y\rangle / \|y\|^2$; then $0 \le \|x - \lambda y\|^2 = \|x\|^2 - |\langle x, y\rangle|^2/\|y\|^2$. Then $\|x+y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Summation by parts: with $F_n = \sum_{k \le n} f_k$ ($F_0 = 0$),

$$\sum_{k=m}^{n} f_k g_k = F_n g_n - F_{m-1} g_m + \sum_{k=m}^{n-1} F_k (g_k - g_{k+1}). \tag{$\dagger$}$$

Theorem 6.18 (6.5.2, Dirichlet–Dedekind test)

Let $f_n, g_n : X \to \mathbb{C}$ satisfy:
(a) the partial sums F_n of $\sum f_n$ are uniformly bounded, $|F_n(x)| \le K$;
(b) $g_n \to 0$ uniformly;
(c) $\sum_n |g_n(x) - g_{n+1}(x)|$ converges uniformly on X. Then $\sum f_n g_n$ converges uniformly on X.

Proof. By ($\dagger$), $|\sum_{k=m}^{n} f_k g_k| \le K|g_n| + K|g_m| + K\sum_{k=m}^{n-1}|g_k - g_{k+1}|$, which is uniformly small for large m; apply Theorem 6.2. ∎

Corollary 6.19 (6.5.3)

If the partial sums of $\sum a_n$ are bounded and b_n decreases to 0, then $\sum a_n b_n$ converges.

Proof. Here $\sum |b_n - b_{n+1}| = b_1$; apply Theorem 6.18 with constant functions. ∎

This contains the alternating series test (Theorem 2.17 of Volume I). For $x \notin 2\pi\mathbb{Z}$,

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\frac{e^{inx} - 1}{e^{ix} - 1}, \qquad \Big|\sum_{k=1}^{n} e^{ikx}\Big| \le \frac{1}{|\sin(x/2)|}, \tag{$\ddagger$}$$

so the partial sums of $\sum \sin kx$ and $\sum \cos kx$ are bounded by $1/\sin(\delta/2)$ on $[\delta, 2\pi - \delta]$.

Theorem 6.20 (6.5.4)

If $a_n \to 0$ and $\sum|a_n - a_{n-1}| < \infty$, then for every $\delta \in (0, \pi)$ the series $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converge uniformly on $[\delta, 2\pi - \delta]$.

Proof. Theorem 6.18 with $f_n(x) = e^{inx}$ (bounded partial sums by (‡)) and constant $g_n = a_n$; take real and imaginary parts. ■

Theorem 6.21 (6.5.5)

Let a_n decrease to 0. Then $\sum a_n \sin nx$ converges uniformly on $\mathbb{R}$ iff $na_n \to 0$.

Proof. ($\Rightarrow$) For large n, $|\sum_{n/2 \le k \le n} a_k \sin kx| < \epsilon$ for all x. At $x = \pi/(2n)$, $kx \in [\pi/4, \pi/2]$ gives $\sin kx \ge 1/\sqrt{2}$, and $a_k \ge a_n$ for these $\ge n/2$ terms, so $\frac{n}{2} a_n/\sqrt{2} < \epsilon$.

($\Leftarrow$) (†), (‡) and $\sin(x/2) \ge x/\pi$ give $|\sum_{k=m}^{n} a_k \sin kx| \le 2a_m/\sin(x/2) \le 2\pi a_m/x$ on $(0, \pi]$. Let $\epsilon_m = \sup_{k \ge m} ka_k \to 0$, $R_m(x) = \sum_{k \ge m} a_k \sin kx$ (by oddness and periodicity, $x \in (0, \pi]$ suffices), and $N = \lfloor \pi/x \rfloor$. If $m > N$: $m > \pi/x$, so $|R_m(x)| \le 2\pi a_m/x < 2ma_m \le 2\epsilon_m$. If $m \le N$: $|\sum_{k=m}^{N}| \le N \cdot \epsilon_m x \le \pi\epsilon_m$ (as $|a_k \sin kx| \le ka_k x$), and $|\sum_{k>N}| \le 2\pi a_{N+1}/x < 2(N+1)a_{N+1} \le 2\epsilon_m$. So $\sup_x |R_m(x)| \le (\pi + 2)\epsilon_m \to 0$. ■

Theorem 6.22 (6.5.6, Abel's test)

Let $f_n : X \to \mathbb{C}$ and $g_n : X \to \mathbb{R}$ satisfy:
(a) $\sum f_n$ converges uniformly;
(b) $g_{n+1}(x) \le g_n(x)$ for all x, n;
(c) (g_n) is uniformly bounded, $|g_n| \le M$. Then $\sum f_n g_n$ converges uniformly.

Proof. By (†) with $R_k = \sum_{j=m}^{k} f_j$, $\sum_{k=m}^{n} f_k g_k = R_n g_n + \sum_{k=m}^{n-1} R_k(g_k - g_{k+1})$. For large m, $|R_k| < \epsilon$ ($k \ge m$) by (a), and $g_k - g_{k+1} \ge 0$, so the sum is $\le \epsilon M + \epsilon(g_m - g_n) \le 3M\epsilon$. ■

6.6 Selected exercises

Motivation. Book exercises: Dini's theorem, termwise integration, a conditional Cauchy product, and Tauber's theorem.

Exercise 6.6.6 (Dini's theorem)

Let $K \subset \mathbb{R}^n$ be compact and let $f_n : K \to [0, \infty)$ be continuous with $f_{n+1} \le f_n$ for all n. Show that if $f_n \to 0$ pointwise on K, then $f_n \to 0$ uniformly on K.

Solution.

Step 1. Fix $\epsilon > 0$; $U_n = \{x \in K : f_n(x) < \epsilon\} = V_n \cap K$ with V_n open (continuity), $U_n \subset U_{n+1}$ ($f_{n+1} \le f_n$), and $K \subset \bigcup_n V_n$ ($f_n \to 0$ pointwise).

Step 2. By compactness $K \subset V_{n_1} \cup \cdots \cup V_{n_r}$, so $K \subset U_N$, $N = \max_j n_j$.

Step 3. For $n \ge N$, $0 \le f_n \le f_N < \epsilon$ on K: $\|f_n\|_{\sup} \le \epsilon$.

No hypothesis can be dropped: x^n on $[0, 1)$ (not compact), $nx(1-x)^n$ on $[0, 1]$ (not monotone; maximum $\to e^{-1}$). ■

Exercise 6.6.12 (A zero from an integral)

Let $f(x)=\sum_{n\ge 0}a_n x^n$ have radius of convergence greater than 1, with real a_n satisfying

$$0<a_0<-\sum_{k=1}^{\infty}\frac{a_{2k}}{2k+1}.$$

Show that f has at least one zero in $(-1,1)$.

Solution.

Step 1. The series converges uniformly on $[-1,1]$ (Theorem 6.12), with $\sum_k|a_{2k}|<\infty$; integrate termwise (Proposition 6.5), using $\int_{-1}^{1}x^n dx=0$ (n odd), $\frac{2}{n+1}$ (n even):

$$\int_{-1}^{1}f(x)\,dx=2\Big(a_0+\sum_{k=1}^{\infty}\frac{a_{2k}}{2k+1}\Big)<0.$$

Step 2. If f had no zero in $(-1,1)$, then $f(0)=a_0>0$ and the intermediate value theorem give $f>0$ there, so $\int_{-1}^{1}f\ge 0$, contradicting Step 1. ∎

Exercise 6.6.13 (Square of the alternating harmonic series)

Let $a_n=\frac{(-1)^{n+1}}{n+1}$ $(n\ge 0)$. Show that the Cauchy product of $\sum a_n$ with itself is

$$2\sum_{m=1}^{\infty}\frac{(-1)^{m+1}}{m+1}\Big(1+\frac{1}{2}+\cdots+\frac{1}{m}\Big),$$

that this series converges, and that its sum is $(\log 2)^2$.

Solution.

Step 1. (The product.) Write $H_m=1+\frac{1}{2}+\cdots+\frac{1}{m}$. For $n\ge 0$,

$$c_n=\sum_{k=0}^{n}a_k a_{n-k}=(-1)^n\sum_{k=0}^{n}\frac{1}{(k+1)(n+1-k)}.$$

By $\frac{1}{(k+1)(n+1-k)}=\frac{1}{n+2}\big(\frac{1}{k+1}+\frac{1}{n+1-k}\big)$ the sum is $\frac{2H_{n+1}}{n+2}$; with $m=n+1$, $c_{m-1}=2(-1)^{m+1}\frac{H_m}{m+1}$.

Step 2. (Convergence.) Let $b_m=\frac{H_m}{m+1}$. Since $H_{m+1}=H_m+\frac{1}{m+1}$,

$$b_m-b_{m+1}=\frac{(m+2)H_m-(m+1)H_{m+1}}{(m+1)(m+2)}=\frac{H_m-1}{(m+1)(m+2)}\ge 0.$$

$H_m\le 1+\log m$ ($\frac{1}{k}\le\int_{k-1}^{k}\frac{dx}{x}$), so $b_m\to 0$; $\sum(-1)^{m+1}b_m$ converges (Corollary 6.19).

Step 3. (The sum.) $\sum a_n=-\log 2$ (Example 6.4) converges only conditionally, so Mertens (Proposition 6.14) fails to apply; but all three series converge, and Corollary 6.16 gives $(\log 2)^2$. ∎

Exercise 6.6.14 (Tauber's theorem)

Let $f(x)=\sum_{n\ge 0}a_n x^n$ converge on $(-1,1)$, and suppose $\lim_{x\to 1^-}f(x)=L$ and $\lim_n na_n=0$. Let $s_n=\sum_{k\le n}a_k$.
(a) Show that $\lim_{n\to\infty}(s_n-f(1-\frac{1}{n}))=0$.
(b) Show that $\sum_{n\ge 0}a_n=L$.

Solution.

(a) Let $x_n = 1 - \frac{1}{n}$ $(n \geq 2)$. Then

$$s_n - f(x_n) = \sum_{k \leq n} a_k (1 - x_n^k) - \sum_{k > n} a_k x_n^k.$$

Step 1. For $0 \leq x < 1$, $1 - x^k = (1 - x)(1 + x + \cdots + x^{k-1}) \leq k(1 - x)$. Hence

$$\Big| \sum_{k \leq n} a_k (1 - x_n^k) \Big| \leq \frac{1}{n} \sum_{k=0}^{n} k|a_k| = \frac{n+1}{n} \cdot \frac{1}{n+1} \sum_{k=0}^{n} k|a_k| \to 0,$$

as Cesàro means of the null sequence $k|a_k|$ tend to 0 (Lemma 9.12).

Step 2. $\epsilon_n = \sup_{k>n} k|a_k| \to 0$ and $|a_k| < \epsilon_n / n$ for $k > n$, so

$$\Big| \sum_{k > n} a_k x_n^k \Big| \leq \frac{\epsilon_n}{n} \sum_{k \geq 0} x_n^k = \frac{\epsilon_n}{n} \cdot \frac{1}{1 - x_n} = \epsilon_n \to 0.$$

Steps 1 and 2 give (a).

(b) $s_n = f(x_n) + (s_n - f(x_n)) \to L$ by (a).

Without $na_n \to 0$ this fails: $a_n = (-1)^n$ gives $f(x) = \frac{1}{1+x} \to \frac{1}{2}$, yet $\sum a_n$ diverges. ■

Chapter 7. Function Spaces

7.1 The space $C(X)$

Motivation. $C(X)$ with the sup norm is complete, and polynomials are dense in it.

A **norm** on a real or complex vector space V is a function $\|\cdot\|$ with $\|v\| \geq 0$ (equality iff $v = 0$), $\|\lambda v\| = |\lambda|\|v\|$, and $\|u + v\| \leq \|u\| + \|v\|$; V is then a **normed space**, with distance $\|u - v\|$, and convergence, Cauchy sequences, open and closed sets are defined as in $\mathbb{R}^n$. A normed space in which every Cauchy sequence converges is **complete** (a **Banach space**). For compact $X \subset \mathbb{R}^n$, $C(X)$ is the space of continuous $f : X \to \mathbb{C}$ with $\|f\|_{\sup} = \max_X |f|$; convergence in this norm is uniform convergence.

Theorem 7.1 (7.1.1)

For compact X, every Cauchy sequence in $C(X)$ converges: $C(X)$ is complete.

Proof. A Cauchy sequence in $C(X)$ is uniformly Cauchy, so converges uniformly (Theorem 6.2) to a function that is continuous (Theorem 6.1). ■

Heine–Borel fails in $C(X)$: $\{x^n\}$ on $[0, 1]$ is closed and bounded, yet no subsequence converges uniformly. The missing ingredient is equicontinuity. A family $\mathcal{F} \subset C(X)$ is **equicontinuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \epsilon$ for all $f \in \mathcal{F}$ and $\|x - y\| < \delta$; it is **bounded** if $\sup_{f \in \mathcal{F}} \|f\|_{\sup} < \infty$.

Proposition 7.2 (7.1.2)

If X is compact and (f_n) converges uniformly in $C(X)$, then $\{f_n\}$ is equicontinuous.

Proof. Choose N with $\|f_n - f_N\|_{\sup} < \epsilon/3$ for $n \geq N$ (Theorem 6.2) and δ working with $\epsilon/3$ for the uniformly continuous $f_1, \dots, f_N$ (Theorem 3.18 of Volume I). For $n > N$, $\|x - y\| < \delta$, $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \epsilon$. ■

Theorem 7.3 (7.1.3, Arzelà–Ascoli)

If X is compact and $\mathcal{F} \subset C(X)$ is bounded and equicontinuous, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Proof. Let $f_n \in \mathcal{F}$, $|f_n| \leq M$. X has a countable dense $E = \{p_1, p_2, \dots\}$ (centres of finitely many $1/k$-balls covering X, all k).

Choose nested subsequences $(f_{k,n})_n$ with $(f_{k,n}(p_k))_n$ convergent; the diagonal $g_n = f_{n,n}$ is eventually a subsequence of each, so $(g_n(p_k))$ converges for every k.

Take δ from equicontinuity for $\epsilon/3$, points $p_{k_1}, \dots, p_{k_r}$ forming a δ-net, and N with $|g_m(p_{k_j}) - g_n(p_{k_j})| < \epsilon/3$ for $m, n \geq N$, all j. For $x \in X$ and $p = p_{k_j}$ with $\|x - p\| < \delta$, $|g_m(x) - g_n(x)| \leq |g_m(x) - g_m(p)| + |g_m(p) - g_n(p)| + |g_n(p) - g_n(x)| < \epsilon$. So (g_n) converges in $C(X)$ (Theorem 7.1). ■

Combined with Proposition 7.2: a subset of $C(X)$ has compact closure iff it is bounded and equicontinuous.

Theorem 7.4 (7.1.4, Weierstrass approximation)

For every continuous $f : [a, b] \to \mathbb{R}$ there are polynomials $P_n \to f$ uniformly on $[a, b]$.

Proof. Take $[a, b] = [0, 1]$ and (subtracting $f(0) + x(f(1) - f(0))$) $f(0) = f(1) = 0$; extended by 0, f is uniformly continuous on $\mathbb{R}$. Let $Q_n(x) = c_n(1 - x^2)^n$ on $[-1, 1]$ with $\int_{-1}^{1} Q_n = 1$. As $(1 - x^2)^n \geq 1 - nx^2$,

$$\int_{-1}^{1} (1 - x^2)^n dx \geq 2 \int_0^{1/\sqrt{n}} (1 - nx^2) dx = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}},$$

so $c_n < \sqrt{n}$, and for $\delta \leq |x| \leq 1$, $Q_n(x) \leq \sqrt{n}(1 - \delta^2)^n \to 0$ uniformly. Let

$$P_n(x) = \int_{-1}^{1} f(x + t) Q_n(t)\, dt = \int_0^1 f(t) Q_n(t - x)\, dt \qquad (0 \leq x \leq 1),$$

a polynomial in x. With $|f(y) - f(x)| < \epsilon/2$ for $|y - x| < \delta$ and $M = \max|f|$,

$$|P_n(x) - f(x)| \leq \int_{-1}^{1} |f(x + t) - f(x)| Q_n(t)\, dt \leq 2M \cdot 2\sqrt{n}(1 - \delta^2)^n + \frac{\epsilon}{2} < \epsilon$$

for large n, uniformly in $x \in [0, 1]$. ∎

For complex-valued f, approximate real and imaginary parts. Consequently $C[a, b]$ has a countable dense set (polynomials with rational coefficients).

7.2 Sequence spaces

Motivation. ℓ^1, ℓ^2: the simplest infinite-dimensional complete normed spaces, models for Chapter 10.

$\ell^1 = \ell^1(\mathbb{N})$ is the space of sequences $a = (a(n))$ with $\|a\|_1 = \sum |a(n)| < \infty$, and ℓ^2 those with $\|a\|_2 = (\sum |a(n)|^2)^{1/2} < \infty$. ℓ^2 is a vector space with inner product $\langle a, b \rangle = \sum a(n)\overline{b(n)}$, which converges absolutely since $|a(n)b(n)| \leq \frac{1}{2}(|a(n)|^2 + |b(n)|^2)$; $\|\cdot\|_2$ is the associated norm (Theorem 6.17).

Theorem 7.5 (7.2.1)

ℓ^1 is complete.

Proof. For Cauchy (a_m), $|a_m(n) - a_k(n)| \leq \|a_m - a_k\|_1$, so $a_m(n) \to a(n)$. If $\|a_m - a_k\|_1 < \epsilon$ for $m, k \geq N$, then $\sum_{n \leq L} |a_m(n) - a_k(n)| < \epsilon$; letting $k \to \infty$, then $L \to \infty$, $\|a_m - a\|_1 \leq \epsilon$ $(m \geq N)$. So $a = a_m - (a_m - a) \in \ell^1$ and $a_m \to a$. ∎

Theorem 7.6 (7.2.2)

ℓ^2 is complete.

Proof. As in Theorem 7.5: the coordinates converge, and $\sum_{n \leq L} |a_m(n) - a_k(n)|^2 < \epsilon^2$ passes to the limit in k and then in L. ∎

Likewise ℓ^∞ (sup norm) is complete. The unit vectors $e_n \in \ell^2$ have $\|e_m - e_n\|_2 = \sqrt{2}$: closed bounded sets in infinite dimensions need not be compact.

7.3 Improper integrals

Motivation. Improper integrals handle infinite intervals and unbounded integrands, with tests parallel to series.

Let I be an interval and $f : I \to \mathbb{C}$ Riemann integrable on every compact subinterval. If $I = [a, b)$ ($b \le \infty$), the **improper integral** $\int_I f = \lim_{c \to b^-} \int_a^c f$ when the limit exists; similarly at a left end point, and for open intervals both ends are treated separately. We write $\int_I |f| < \infty$ when this improper integral converges (**absolute convergence**).

Proposition 7.7 (7.3.1)

If $0 \le f \le g$ on I and $\int_I g$ converges, then $\int_I f$ converges.

Proof. $c \mapsto \int_a^c f$ is increasing and bounded by $\int_I g$, so it has a limit as $c \to b^-$ (as for monotone sequences). ■

Proposition 7.8 (7.3.2, comparison test)

If $|f| \le g$ and $\int_I g$ converges, then $\int_I |f|$ and $\int_I f$ converge and $|\int_I f| \le \int_I |f| \le \int_I g$.

Proof. $\int_I |f|$ converges by Proposition 7.7. Cauchy criterion: for $a < c < c' < b$, $|\int_c^{c'} f| \le \int_c^{c'} |f| \to 0$ as $c, c' \to b$, so $\lim \int_a^c f$ exists (as in Theorem 2.14 of Volume I). The inequality passes to the limit from finite intervals (Theorem 5.5 of Volume I). ■

Example 7.1

$\int_1^\infty x^{-p} dx < \infty$ iff $p > 1$, and $\int_0^1 x^{-p} dx < \infty$ iff $p < 1$. $\int_0^\infty \frac{\sin x}{x}\, dx$ converges (integrate by parts: $\int_1^c \frac{\sin x}{x} dx = \frac{\cos 1}{1} - \frac{\cos c}{c} - \int_1^c \frac{\cos x}{x^2} dx$) but not absolutely (on $[k\pi, (k+1)\pi]$, $\int |\sin x|/x \ge \frac{2}{(k+1)\pi}$).

Proposition 7.9 (7.3.3, integral test)

If $f : [1, \infty) \to (0, \infty)$ is decreasing, then $\sum f(n)$ and $\int_1^\infty f$ converge or diverge together; indeed $\sum_{n=2}^N f(n) \le \int_1^N f \le \sum_{n=1}^{N-1} f(n)$, and if $f \to 0$, then $\sum_{n \le N} f(n) - \int_1^N f$ converges.

Proof. On $[n, n+1]$, $f(n+1) \le f \le f(n)$; integrate and sum. The differences $d_N = \sum_{n \le N} f(n) - \int_1^N f$ decrease ($d_N - d_{N+1} = \int_N^{N+1} f - f(N+1) \ge 0$) and are $\ge f(N) > 0$, so converge. ■

For $f(x) = 1/x$ this gives Euler's constant $\gamma = \lim(1 + \frac{1}{2} + \cdots + \frac{1}{N} - \log N)$.

Theorem 7.10 (7.3.4)

If $\int_I |f| < \infty$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $\int_x^y |f| < \epsilon$ whenever $x < y$ in I and $y - x < \delta$.

Proof. Choose a compact $[c, d] \subset I$ with $\int_{I - [c,d]} |f| < \epsilon/2$; f is bounded on $[c, d]$, say by K. With $\delta = \epsilon/(2K + 1)$, $\int_x^y |f| \le \int_{[x,y] \cap [c,d]} |f| + \int_{I-[c,d]} |f| < K\delta + \epsilon/2 < \epsilon$. ■

7.4 Spaces of integrable functions

Motivation. Integral norms on functions; their incompleteness under Riemann integration motivates Chapter 10.

Proposition 7.11 (7.4.1)

For a normed space X the following are equivalent:
(a) every Cauchy sequence in X converges;
(b) whenever $\sum\|x_n\| < \infty$, the series $\sum x_n$ converges in X.

Proof. (a)⇒(b): the partial sums are Cauchy, since $\|\sum_{m<k\le n} x_k\| \le \sum_{m<k\le n}\|x_k\|$.

(b)⇒(a): given a Cauchy (y_n), choose $n_1 < n_2 < \cdots$ with $\|y_m - y_n\| < 2^{-k}$ for $m, n \ge n_k$. With $x_1 = y_{n_1}$, $x_{k+1} = y_{n_{k+1}} - y_{n_k}$, $\sum\|x_k\| < \infty$, so $y_{n_k} = \sum_{j\le k} x_j$ converges; a Cauchy sequence with a convergent subsequence converges (Lemma 2.13 of Volume I, same proof). ∎

Let $\mathcal{R}^1(I)$ and $\mathcal{R}^2(I)$ be the locally Riemann integrable $f : I \to \mathbb{C}$ with $\int_I|f| < \infty$, resp. $\int_I|f|^2 < \infty$, with $\|f\|_1 = \int_I|f|$ and $\|f\|_2 = (\int_I|f|^2)^{1/2}$. These are seminorms ($\|f\| = 0 \not\Rightarrow f = 0$); $\mathcal{R}^2$ has inner product $\langle f, g\rangle = \int_I f\bar{g}$. Neither space is complete (hence Chapter 10). A **step function** is a finite linear combination of characteristic functions of bounded intervals.

Lemma 7.12 (7.4.2)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_2 < \epsilon$.

Proof. Let f be real (else split). Choose $[a, b] \subset I$ with $\int_{I-[a,b]}|f|^2 < \epsilon^2/4$, $|f| \le K$ on $[a, b]$, a partition with $U(f, P) - L(f, P) < \epsilon^2/(8K)$, and $s = m_i$ on the i-th interval (0 off $[a, b]$). Then $0 \le f - s \le 2K$ on $[a, b]$, so $\int_a^b|f - s|^2 \le 2K\int_a^b(f - s) \le 2K(U - L) < \epsilon^2/4$. Altogether $\|f - s\|_2^2 < \epsilon^2/2$. ∎

Lemma 7.13 (7.4.3)

For a step function s on I and $\epsilon > 0$ there is a continuous g with $\|s - g\|_2 < \epsilon$ (and $g = 0$ outside a compact subinterval).

Proof. Take $s = \chi_{[a,b]}$ (any end points) and $g = 1$ on $[a + \eta, b - \eta]$, 0 off (a, b), linear between; $|s - g| \le 1$ is nonzero only on two intervals of length η, so $\|s - g\|_2^2 \le 2\eta$. ∎

Theorem 7.14 (7.4.4)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a continuous g with $\|f - g\|_2 < \epsilon$.

Proof. Combine Lemmas 7.12 and 7.13 with the triangle inequality for $\|\cdot\|_2$. ∎

Corollary 7.15 (7.4.5)

For $f \in \mathcal{R}^2[a, b]$ there are polynomials P_n with $\|f - P_n\|_2 \to 0$.

Proof. Choose continuous g with $\|f - g\|_2 < \epsilon$ and a polynomial P with $\|g - P\|_{\sup} < \epsilon$ (Theorem 7.4); then $\|g - P\|_2 \le \epsilon\sqrt{b - a}$. ∎

Theorem 7.16 (7.4.6)

For $f \in \mathcal{R}^1(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_1 < \epsilon$, and a continuous g vanishing outside a compact subinterval with $\|f - g\|_1 < \epsilon$. For $f \in \mathcal{R}^1[a,b]$ there are polynomials P with $\|f - P\|_1 < \epsilon$.

Proof. As in Lemma 7.12, with $\int_a^b |f - s| = \int_a^b (f - s) \le U - L$ for the lower step function. For the continuous approximation of $\chi_{[a,b]}$ the error in $\|\cdot\|_1$ is at most η. Polynomials: $\|g - P\|_1 \le (b-a)\|g - P\|_{\sup}$. ∎

7.5 Selected exercises

Motivation. Book exercises: moments, Bernstein polynomials, and compactness in ℓ^2.

Exercise 7.5.1 (Moments determine a function)

Let $f : [0,1] \to \mathbb{R}$ be continuous with $\int_0^1 f(x) x^n \, dx = 0$ for $n = 0, 1, 2, \ldots$. Show that $f = 0$.

Solution.

Step 1. By linearity, $\int_0^1 fP = 0$ for every polynomial P.

Step 2. Take polynomials $P_n \to f$ uniformly (Theorem 7.4):

$$\int_0^1 f^2 = \int_0^1 f(f - P_n) \le \|f\|_{\sup} \|f - P_n\|_{\sup} \to 0,$$

so $\int_0^1 f^2 = 0$.

Step 3. If $f(x_0) \neq 0$, then $f^2 > \tfrac{1}{2} f(x_0)^2$ on an interval of length $\eta > 0$, so $\int_0^1 f^2 \ge \tfrac{1}{2} f(x_0)^2 \eta > 0$, a contradiction. ∎

Exercise 7.5.5 (Weights concentrating at an end point)

Let $f : [0,1] \to \mathbb{R}$ be continuous. Show that

$$\lim_{n \to \infty} \frac{\int_0^1 x^n f(x) \, dx}{\int_0^1 x^n \, dx} = f(1).$$

Solution.

Step 1. As $\int_0^1 x^n dx = \frac{1}{n+1}$, the quotient minus $f(1)$ is $(n+1) \int_0^1 x^n \big(f(x) - f(1)\big) dx$.

Step 2. With $M = \max |f|$ and $|f(x) - f(1)| < \epsilon$ on $[1 - \delta, 1]$,

$$(n+1) \int_0^{1-\delta} x^n |f(x) - f(1)| \, dx \le 2M(1-\delta)^{n+1},$$
$$(n+1) \int_{1-\delta}^1 x^n |f(x) - f(1)| \, dx \le \epsilon \, (n+1) \int_0^1 x^n dx = \epsilon.$$

Step 3. The first bound $\to 0$, so the quantity in Step 1 is $< 2\epsilon$ for large n. ∎

Exercise 7.5.7 (Bernstein polynomials)

Let $f:[0,1]\to\mathbb{R}$ be continuous, $r_k(x)=\binom{n}{k}x^k(1-x)^{n-k}$, and $p_n(x)=\sum_{k=0}^n\binom{n}{k}f(\frac{k}{n})x^k(1-x)^{n-k}$.
(a) Show that $\sum_{k=0}^n(k-nx)^2r_k(x)=nx(1-x)$.
(b) Show that $p_n\to f$ uniformly on $[0,1]$.

Solution.

(a)

Step 1. By the binomial theorem, $\sum_k r_k(x)=(x+(1-x))^n=1$.

Step 2. From $k\binom{n}{k}=n\binom{n-1}{k-1}$ and $k(k-1)\binom{n}{k}=n(n-1)\binom{n-2}{k-2}$, the binomial theorem again gives

$$\sum_k k\,r_k(x)=nx,\qquad \sum_k k(k-1)\,r_k(x)=n(n-1)x^2.$$

Step 3. Expanding $(k-nx)^2=k(k-1)+k-2nxk+n^2x^2$,

$$\sum_k(k-nx)^2r_k(x)=n(n-1)x^2+nx-2n^2x^2+n^2x^2=nx(1-x).$$

(b)

Step 1. By (a), Step 1, $f(x)-p_n(x)=\sum_k(f(x)-f(\frac{k}{n}))r_k(x)$. Let $M=\max|f|$ and $|f(x)-f(y)|<\epsilon$ for $|x-y|<\delta$.

Step 2. Indices with $|\frac{k}{n}-x|<\delta$ contribute $\le\epsilon\sum_k r_k(x)=\epsilon$.

Step 3. The others have $(k-nx)^2\ge n^2\delta^2$, so by (a) contribute at most

$$2M\sum_k\frac{(k-nx)^2}{n^2\delta^2}r_k(x)=\frac{2Mx(1-x)}{n\delta^2}\le\frac{M}{2n\delta^2}.$$

Step 4. So $\|f-p_n\|_{\sup}\le\epsilon+\frac{M}{2n\delta^2}<2\epsilon$ for large n: a constructive proof of Theorem 7.4. ■

Exercise 7.5.9 (Compact and non-compact boxes in ℓ^2)

Show that $K=\{a\in\ell^2:|a(n)|\le\frac{1}{n},\ n=1,2,\dots\}$ is compact, i.e. every sequence in K has a subsequence converging in ℓ^2 to a point of K. Is $L=\{a\in\ell^2:|a(n)|\le\frac{1}{\sqrt{n}},\ n=1,2,\dots\}$ compact?

Solution.

(a) K is compact.

Step 1. For (a_m) in K, $|a_m(n)|\le\frac{1}{n}$; the diagonal argument of Theorem 7.3 gives a subsequence (b_j) with $b_j(n)\to a(n)$ for all n.

Step 2. $|a(n)|\le\frac{1}{n}$, so $a\in K$.

Step 3. Choose N with $\sum_{n>N}\frac{4}{n^2}<\frac{\epsilon^2}{2}$ and j_0 with $\sum_{n\le N}|b_j(n)-a(n)|^2<\frac{\epsilon^2}{2}$ $(j\ge j_0)$. As $|b_j(n)-a(n)|\le\frac{2}{n}$,

$$\|b_j-a\|_2^2=\sum_{n\le N}|b_j(n)-a(n)|^2+\sum_{n>N}|b_j(n)-a(n)|^2<\epsilon^2\qquad(j\ge j_0).$$

(b) L is not compact.

Step 1. For $m \geq 1$ let $c_m(n) = \frac{1}{\sqrt{n}}$ for $2^m < n \leq 2^{m+1}$ and $c_m(n) = 0$ otherwise. Then $c_m \in L$ and

$$\|c_m\|_2^2 = \sum_{2^m < n \leq 2^{m+1}} \frac{1}{n} \geq 2^m \cdot \frac{1}{2^{m+1}} = \frac{1}{2}.$$

Step 2. Disjoint supports give $\|c_m - c_{m'}\|_2^2 \geq 1$ $(m \neq m')$: no subsequence is Cauchy. ∎

Chapter 8. Functions Defined by Integrals

8.1 Integrals depending on a parameter

Motivation. When is $x \mapsto \int f(x,t)\,dt$ continuous, and differentiable under the integral sign?

For f on $[a,b]\times[c,d]$ let $\phi(x)=\int_c^d f(x,t)\,dt$. Write $D_1 f=\partial f/\partial x$.

Proposition 8.1 (8.1.1)

Let $f:[a,b]\times[c,d]\to\mathbb{R}$ be continuous.
(a) ϕ is continuous on $[a,b]$.
(b) If $D_1 f$ exists and is continuous on $(a,b)\times[c,d]$, then ϕ is differentiable on (a,b) and $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. (a) By uniform continuity (Theorem 3.18 of Volume I), $|f(x,t)-f(x',t)|<\epsilon$ for $|x-x'|<\delta$, all t, so $|\phi(x)-\phi(x')|\le\epsilon(d-c)$.

(b) $D_1 f$ is uniformly continuous on $[x-\eta,x+\eta]\times[c,d]\subset(a,b)\times[c,d]$. For $0<|h|<\eta$, the mean value theorem gives $\frac{f(x+h,t)-f(x,t)}{h}=D_1 f(x+\theta h,t)$ with $0<\theta=\theta(h,t)<1$, so

$$\Big|\frac{\phi(x+h)-\phi(x)}{h}-\int_c^d D_1 f(x,t)\,dt\Big|\le\int_c^d |D_1 f(x+\theta h,t)-D_1 f(x,t)|\,dt\le\epsilon(d-c)$$

once $|h|$ is small. ∎

Example 8.1

Differentiating $\int_0^1 t^x\,dt=\frac{1}{x+1}$ n times in $x>-1$ gives $\int_0^1 t^x(\log t)^n dt=(-1)^n n!/(x+1)^{n+1}$. Also $\int_a^b\phi(x)\,dx=\int_c^d\int_a^b f(x,t)\,dx\,dt$ for continuous f: as functions of b both vanish at $b=a$ with derivative $\int_c^d f(b,t)\,dt$ ((a), (b), Theorem 5.11 of Volume I).

For f continuous on $[a,b]\times[c,d)$ $(d\le\infty)$, consider the improper integral $\phi(x)=\int_c^d f(x,t)\,dt$.

Theorem 8.2 (8.1.2)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, and suppose there is $g:[c,d)\to\mathbb{R}$ with (a) $|f(x,t)|\le g(t)$ for all (x,t) and (b) $\int_c^d g<\infty$. Then $\int_c^d f(x,t)\,dt$ converges for each x and ϕ is continuous on $[a,b]$.

Proof. Convergence: comparison (Proposition 7.8). If $\int_{d'}^d g<\epsilon$, then $|\phi-\phi_{d'}|<\epsilon$ with $\phi_{d'}(x)=\int_c^{d'} f(x,t)\,dt$ continuous (Proposition 8.1); so ϕ is a uniform limit of continuous functions (Theorem 6.1). ∎

Theorem 8.3 (8.1.3)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, $D_1 f$ exist and be continuous on $(a,b)\times[c,d)$, and $\int_c^d f(x_0,t)\,dt$ converge for some $x_0\in(a,b)$. If there is $h:[c,d)\to\mathbb{R}$ with (a) $|D_1 f(x,t)|\le h(t)$ and (b) $\int_c^d h<\infty$, then $\int_c^d f(x,t)\,dt$ converges for each $x\in(a,b)$, and ϕ is differentiable with $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. $|f(x,t)-f(x_0,t)| \le |x-x_0|h(t)$ (mean value theorem) gives convergence by comparison. $\phi_{d'}(x) = \int_c^{d'} f(x,t)dt$ has $\phi'_{d'}(x) = \int_c^{d'} D_1 f(x,t)dt$ (Proposition 8.1), converging uniformly as $d' \to d$ (tails $\le \int_{d'}^{d} h$), and $\phi_{d'}(x_0) \to \phi(x_0)$; apply Theorem 6.4 along $d'_n \to d$. ∎

Example 8.2

(a) $F(x) = \int_0^\infty e^{-t^2}\cos(xt)\,dt$ satisfies $F'(x) = -\int_0^\infty te^{-t^2}\sin(xt)\,dt = -\frac{x}{2}F(x)$ (integration by parts), so $F(x) = F(0)e^{-x^2/4} = \frac{\sqrt{\pi}}{2}e^{-x^2/4}$.
(b) $\int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}$: for $G(s) = \int_0^\infty e^{-sx}\frac{\sin x}{x}dx$ $(s>0)$, Theorem 8.3 gives $G'(s) = -\frac{1}{1+s^2}$, and $G(s) \to 0$ as $s \to \infty$, so $G(s) = \frac{\pi}{2} - \arctan s$; continuity at $s=0$ follows from the second mean value theorem (Corollary 5.25 of Volume I).

8.2 The gamma function

Motivation. Γ extends the factorial; Section 8.1 gives its basic properties.

For $x>0$ let $\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,dt$, convergent since $t^{x-1}e^{-t} \le t^{x-1}$ near 0 and $\le Ct^{-2}$ near ∞. By Theorem 8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(n)}(x) = \int_0^\infty t^{x-1}(\log t)^n e^{-t}dt$. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$, and $\Gamma(1)=1$, so $\Gamma(n+1) = n!$.

A function $f: I \to \mathbb{R}$ is **convex** if $f((1-\lambda)x+\lambda y) \le (1-\lambda)f(x)+\lambda f(y)$ for $x,y \in I$, $0 \le \lambda \le 1$.

Proposition 8.4 (8.2.1)

(a) A continuous $f: I \to \mathbb{R}$ with $f(\frac{x+y}{2}) \le \frac{f(x)+f(y)}{2}$ for all $x,y \in I$ is convex.
(b) A convex f satisfies, for $s<t<u$ in I,

$$\frac{f(t)-f(s)}{t-s} \le \frac{f(u)-f(s)}{u-s} \le \frac{f(u)-f(t)}{u-t}.$$

Proof. (a) Induct on n for dyadic $\lambda = k/2^n$: if $\lambda = \frac{\lambda_1+\lambda_2}{2}$ with λ_i of level $n-1$, $(1-\lambda)x+\lambda y$ is the midpoint of the points for λ_1, λ_2. General λ: dyadic $\lambda_n \to \lambda$ and continuity.

(b) $t = (1-\lambda)s+\lambda u$, $\lambda = \frac{t-s}{u-s}$, so $f(t) \le (1-\lambda)f(s)+\lambda f(u)$; rearrange. ∎

$\log\Gamma$ is convex: by Hölder's inequality with exponents $1/\lambda$, $1/(1-\lambda)$ (or Cauchy–Schwarz for the midpoint and Proposition 8.4(a)),

$$\Gamma\Big(\frac{x+y}{2}\Big) = \int_0^\infty t^{\frac{x-1}{2}}e^{-t/2}\,t^{\frac{y-1}{2}}e^{-t/2}\,dt \le \Gamma(x)^{1/2}\Gamma(y)^{1/2}.$$

Theorem 8.5 (8.2.2, Bohr–Mollerup)

If $f:(0,\infty)\to(0,\infty)$ satisfies (a) $f(x+1) = xf(x)$, (b) $f(1)=1$, (c) $\log f$ is convex, then $f = \Gamma$.

Proof. Γ has (a)–(c), so we show they determine f, and by (a) on $(0,1]$ suffices. Let $\phi = \log f$, $0<x\le 1$; $\phi(n+1) = \log n!$. Proposition 8.4(b) at $n < n+1 < n+1+x$ and $n+1 < n+1+x < n+2$ gives

$$\log n \le \frac{\phi(n+1+x)-\phi(n+1)}{x} \le \log(n+1).$$

By (a), $\phi(n+1+x) = \phi(x) + \log[x(x+1)\cdots(x+n)]$. Hence

$$0 \le \phi(x) - \log\frac{n!\,n^x}{x(x+1)\cdots(x+n)} \le x\log\Big(1+\frac{1}{n}\Big) \to 0,$$

so $f(x) = \lim_n \frac{n!\,n^x}{x(x+1)\cdots(x+n)}$ (Gauss's formula) is determined. ■

Example 8.3

The **beta function** $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ satisfies $B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$: for fixed y, $f(x) = \Gamma(x+y)B(x,y)/\Gamma(y)$ satisfies (a), (b), (c) of Theorem 8.5. In particular $\Gamma(\frac{1}{2})^2 = B(\frac{1}{2},\frac{1}{2}) = \pi$, so $\int_{-\infty}^{\infty} e^{-x^2}dx = \Gamma(\frac{1}{2}) = \sqrt{\pi}$. Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\,x^x e^{-x}$ follows from the substitution $t = x(1+u)$ and Laplace's method.

8.3 Selected exercises

Motivation. Book exercises: classical integrals by differentiation under the integral sign, and Γ.

Exercise 8.4.1 (The Gaussian integral)

Let
$$f(x) = \Big(\int_0^x e^{-t^2}dt\Big)^2, \qquad g(x) = \int_0^1 \frac{e^{-x^2(t^2+1)}}{t^2+1}\,dt.$$
Compute $(f+g)'$, show that $f(x)+g(x) = \frac{\pi}{4}$, and deduce $\lim_{A\to\infty}\int_0^A e^{-t^2}dt = \frac{1}{2}\sqrt{\pi}$.

Solution.

Step 1. By the fundamental theorem of calculus, $f'(x) = 2e^{-x^2}\int_0^x e^{-t^2}dt$.

Step 2. $h(x,t) = e^{-x^2(t^2+1)}/(t^2+1)$ and $D_1h = -2xe^{-x^2(t^2+1)}$ are continuous, so by Proposition 8.1(b)
$$g'(x) = -2e^{-x^2}\int_0^1 xe^{-x^2t^2}dt = -2e^{-x^2}\int_0^x e^{-u^2}du,$$
($u = xt$, any real x).

Step 3. So $(f+g)' = 0$ and $f+g \equiv f(0)+g(0) = \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}$.

Step 4. $0 \le g(x) \le \frac{\pi}{4}e^{-x^2} \to 0$, so $f(x) \to \frac{\pi}{4}$; taking square roots, $\int_0^A e^{-t^2}dt \to \frac{\sqrt{\pi}}{2}$. ■

Exercise 8.4.4 (A Laplace-type integral)

For $x \in \mathbb{R}$ let
$$f(x) = \int_0^\infty \frac{\cos tx}{1+t^2}\,dt, \qquad g(x) = \int_0^\infty \frac{\sin tx}{t(1+t^2)}\,dt.$$
(a) Show that g is differentiable on $\mathbb{R}$ and $g' = f$.
(b) Show that f is continuous on $\mathbb{R}$.
(c) Show that f is differentiable on $(0,\infty)$ and $g''(x) - g(x) + \frac{\pi}{2} = 0$ for $x > 0$.
(d) Show that $f(x) = \frac{\pi}{2}e^{-|x|}$ for every $x \in \mathbb{R}$.

Solution.

(a) $k(x,t) = \frac{\sin tx}{t(1+t^2)}$ $(t>0)$, $k(x,0) = x$, is continuous ($\frac{\sin tx}{t} \to x$ locally uniformly), $|k| \le \frac{|x|}{1+t^2}$, and $D_1k = \frac{\cos tx}{1+t^2}$ is continuous with $|D_1k| \le \frac{1}{1+t^2}$ integrable. Theorem 8.3 gives $g'(x) = \int_0^\infty \frac{\cos tx}{1+t^2}dt = f(x)$.

(b) $\big|\frac{\cos tx}{1+t^2}\big| \le \frac{1}{1+t^2}$: f is continuous (Theorem 8.2), $|f| \le \frac{\pi}{2} = f(0)$.

(c) Fix $x > 0$.

Step 1. $\frac{1}{t(1+t^2)} = \frac{1}{t} - \frac{t}{1+t^2}$ and $\int_0^\infty \frac{\sin tx}{t}\,dt = \frac{\pi}{2}$ (Example 8.2(b), $u = tx$), so $\int_0^\infty \frac{t\sin tx}{1+t^2}dt$ converges, and $u = tx$ gives

$$g(x) = \frac{\pi}{2} - H(x), \qquad H(x) = \int_0^\infty \frac{u\sin u}{x^2+u^2}\,du.$$

Step 2. Integrating by parts, with $\frac{d}{du}\frac{u}{x^2+u^2} = \frac{x^2-u^2}{(x^2+u^2)^2}$ and boundary terms $-\frac{u\cos u}{x^2+u^2}\big|_0^\infty = 0$,

$$H(x) = \int_0^\infty \frac{(x^2-u^2)\cos u}{(x^2+u^2)^2}\,du.$$

Step 3. Also $f(x) = \int_0^\infty \frac{x\cos u}{x^2+u^2}\,du$, and $\left|\frac{\partial}{\partial x}\frac{x\cos u}{x^2+u^2}\right| = \left|\frac{(u^2-x^2)\cos u}{(x^2+u^2)^2}\right| \le \frac{1}{a^2+u^2}$ for $x \ge a > 0$. Theorem 8.3 on each (a,b) and Step 2 give

$$f'(x) = \int_0^\infty \frac{(u^2-x^2)\cos u}{(x^2+u^2)^2}\,du = -H(x) = g(x) - \frac{\pi}{2}.$$

Step 4. With $g' = f$ from (a), $g''(x) = f'(x) = g(x) - \frac{\pi}{2}$.

(d)

Step 1. $y = g - \frac{\pi}{2}$ has $y'' = y$ on $(0,\infty)$; $w = y' + y$ has $w' = w$, so $w = Ce^x$, $(e^x y)' = Ce^{2x}$, and $y = Ae^x + Be^{-x}$.

Step 2. $f = y' = Ae^x - Be^{-x}$ is bounded (b), so $A = 0$; continuity at 0 and $f(0) = \frac{\pi}{2}$ give $-B = \frac{\pi}{2}$.

Step 3. So $f(x) = \frac{\pi}{2}e^{-x}$ for $x \ge 0$, and f is even. ■

Exercise 8.4.6 (Gamma at half-integers)

Show that $\Gamma\left(n+\frac{1}{2}\right) = \frac{(2n)!}{n!\,4^n}\sqrt{\pi}$ for $n = 0, 1, 2, \ldots$.

Solution. For $n = 0$ this is $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (Example 8.3). If it holds for n, then $\Gamma(x+1) = x\Gamma(x)$ gives

$$\Gamma\Big(n+\frac{3}{2}\Big) = \frac{2n+1}{2}\cdot\frac{(2n)!}{n!\,4^n}\sqrt{\pi} = \frac{(2n+2)(2n+1)(2n)!}{4(n+1)\,n!\,4^n}\sqrt{\pi} = \frac{(2n+2)!}{(n+1)!\,4^{n+1}}\sqrt{\pi},$$

which is the claim for $n+1$. ■

Exercise 8.4.7 (A series for Γ)

Show that for every $x > 0$,

$$\Gamma(x) = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\cdot\frac{1}{n+x} + \sum_{n=0}^{\infty}\Big(\frac{1}{n!}\int_1^\infty t^{-1}e^{-t}(\log t)^n dt\Big)x^n.$$

Solution. Split $\Gamma(x) = \int_0^1 t^{x-1}e^{-t}dt + \int_1^\infty t^{x-1}e^{-t}dt$.

Step 1. (Over $(0,1]$.) $R_N(t) = e^{-t} - \sum_{n\le N}\frac{(-t)^n}{n!}$ has $|R_N| \le \sum_{n>N}\frac{1}{n!} =: \rho_N \to 0$ on $[0,1]$; with $\int_0^1 t^{n+x-1}dt = \frac{1}{n+x}$,

$$\Big|\int_0^1 t^{x-1}e^{-t}dt - \sum_{n\le N}\frac{(-1)^n}{n!(n+x)}\Big| = \Big|\int_0^1 t^{x-1}R_N(t)\,dt\Big| \le \frac{\rho_N}{x} \to 0.$$

Step 2. (Over $[1,\infty)$.) $t^{x-1}e^{-t} = t^{-1}e^{-t}e^{x\log t} = \sum_n \phi_n(t)$, $\phi_n(t) = \frac{x^n}{n!}t^{-1}e^{-t}(\log t)^n \ge 0$, each $\int_1^\infty \phi_n \le \int_1^\infty t^{x-1}e^{-t}$ finite.

Step 3. (Upper bound.) $\sum_{n\le N}\phi_n \le t^{x-1}e^{-t}$, so $\sum_n \int_1^\infty \phi_n \le \int_1^\infty t^{x-1}e^{-t}dt$.

Step 4. (Lower bound.) On $[1,c]$, $\log t \le \log c$, so $\sum \phi_n$ converges uniformly (Corollary 6.3) and (Proposition 6.5)

$$\int_1^c t^{x-1}e^{-t}dt = \sum_n \int_1^c \phi_n \le \sum_n \int_1^\infty \phi_n.$$

Let $c\to\infty$: with Step 3, $\int_1^\infty t^{x-1}e^{-t}dt = \sum_n \int_1^\infty \phi_n$, the second series. Add Step 1. ■

Chapter 9. Fourier Series

9.1 Fourier coefficients

Motivation. Fourier coefficients give the best L^2 approximation by trigonometric polynomials.

In an inner product space X, a family $\{\phi_n : n \in \mathbb{Z}\}$ is **orthonormal** if $\langle \phi_m, \phi_n \rangle = \delta_{mn}$. In $\mathcal{R}^2[-\pi,\pi]$ with $\langle f, g\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\bar{g}$, the functions $e_n(x) = e^{inx}$ are orthonormal. For $f \in \mathcal{R}^1[-\pi,\pi]$ (extended 2π-periodically) the **Fourier coefficients** are $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt$, the **Fourier series** is $\sum \hat{f}(n)e^{inx}$, and $s_n(f;x) = \sum_{|k|\le n} \hat{f}(k)e^{ikx}$.

Theorem 9.1 (9.1.1, best approximation and Bessel's inequality)

Let $\{\phi_n\}$ be orthonormal in X, $x \in X$, $c_n = \langle x, \phi_n\rangle$. For all complex z_n,

$$\Big\| x - \sum_{n=-N}^{N} c_n\phi_n \Big\| \le \Big\| x - \sum_{n=-N}^{N} z_n\phi_n \Big\|,$$

with equality iff $z_n = c_n$ for all $|n| \le N$. Moreover $\sum_{n\in\mathbb{Z}} |c_n|^2 \le \|x\|^2$.

Proof. Let $s = \sum_{|n|\le N} c_n\phi_n$ and $t = \sum z_n\phi_n$. Then $\langle x - s, \phi_n\rangle = 0$ for $|n| \le N$, so $x - s \perp s - t$, and $\|x-t\|^2 = \|x-s\|^2 + \|s-t\|^2 = \|x-s\|^2 + \sum |c_n - z_n|^2$. With $t = 0$: $\|x\|^2 = \|x-s\|^2 + \sum_{|n|\le N}|c_n|^2$, which gives Bessel's inequality. ∎

Proposition 9.2 (9.1.2)

For $f \in \mathcal{R}^1[-\pi,\pi]$:
(a) if f is real, $\hat{f}(-n) = \overline{\hat{f}(n)}$;
(b) if f is even, $\hat{f}(-n) = \hat{f}(n)$;
(c) if f is odd, $\hat{f}(-n) = -\hat{f}(n)$.

Proof. (a) Conjugate the integral. (b), (c) Substitute $t \mapsto -t$. ∎

In real form,

$$s_n(f;x) = \frac{a_0}{2} + \sum_{k\le n} (a_k \cos kx + b_k \sin kx)$$

with $a_k = \frac{1}{\pi}\int f(t)\cos kt\,dt$, $b_k = \frac{1}{\pi}\int f(t)\sin kt\,dt$; even functions have cosine series and odd functions sine series.

Example 9.1

$f(x) = x$ on $(-\pi,\pi)$: $\hat{f}(n) = \frac{(-1)^n i}{n}$ $(n \ne 0)$, so $x \sim 2\sum (-1)^{n+1}\frac{\sin nx}{n}$. $f(x) = |x|$: $|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k \text{ odd}} \frac{\cos kx}{k^2}$, and at $x = 0$ this gives $\sum_{k \text{ odd}} k^{-2} = \pi^2/8$, hence $\sum n^{-2} = \pi^2/6$ (once convergence to f is proved, Corollary 9.7).

9.2 Pointwise convergence

Motivation. The Fourier series converges to f where f is reasonably smooth: Dirichlet kernel plus Riemann–Lebesgue.

Theorem 9.3 (9.2.1, Riemann–Lebesgue lemma)

If $\int_I |f| < \infty$ on an interval I, then for every $\beta \in \mathbb{R}$, $\lim_{\alpha\to\infty} \int_I f(t)\sin(\alpha t+\beta)\,dt = 0$.

Proof. For $f = \chi_{[a,b]}$, $|\int_a^b \sin(\alpha t+\beta)dt| \le 2/\alpha$; by linearity the claim holds for step functions. For general f and ϵ, choose a step function s with $\int_I |f-s| < \epsilon$ (Theorem 7.16); then $|\int f\sin(\alpha t+\beta)| \le \epsilon + |\int s\sin(\alpha t+\beta)| < 2\epsilon$ for large α. ■

In particular $\hat{f}(n) \to 0$ as $|n| \to \infty$ (Theorem 9.21). The partial sums are integrals against the **Dirichlet kernel**:

$$s_n(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_n(t)\,dt, \qquad D_n(t) = \sum_{|k|\le n} e^{ikt} = \frac{\sin(n+\frac{1}{2})t}{\sin\frac{t}{2}}, \qquad \frac{1}{2\pi}\int_{-\pi}^{\pi} D_n = 1.$$

Since D_n is even, $s_n(f;x) = \frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)]D_n(t)\,dt$.

Theorem 9.4 (9.2.2, Riemann localization)

Let $0 < \delta < \pi$ and $\int_{-\pi}^{\pi}|f| < \infty$. Then $\lim_n s_n(f;x)$ exists iff $\lim_n \frac{1}{\pi}\int_0^{\delta}[f(x+t)+f(x-t)]\frac{\sin(n+\frac{1}{2})t}{t}\,dt$ exists, and then they are equal. So convergence of the Fourier series at x depends only on f near x.

Proof. $\frac{1}{\sin(t/2)} - \frac{2}{t}$ is bounded on $(0,\pi]$, so by Theorem 9.3, $\frac{1}{2\pi}\int_0^{\pi}[f(x+t)+f(x-t)](\frac{1}{\sin(t/2)} - \frac{2}{t})\sin(n+\frac{1}{2})t\,dt \to 0$; and $\frac{2}{t}$ is bounded on $[\delta,\pi]$, so the integral over $[\delta,\pi]$ with kernel $\frac{2\sin(n+\frac{1}{2})t}{t}$ also tends to 0. ■

We use $\lim_{\lambda\to\infty}\int_0^{\delta}\frac{\sin\lambda t}{t}\,dt = \lim\int_0^{\lambda\delta}\frac{\sin u}{u}du = \frac{\pi}{2}$ (Example 8.2).

Lemma 9.5 (9.2.3)

If $g(0+)$ exists and $\int_0^{\delta}\frac{|g(t)-g(0+)|}{t}\,dt < \infty$, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^{\delta} g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. Write $g(t) = g(0+) + (g(t)-g(0+))$. The first part gives $g(0+)\cdot\frac{2}{\pi}\cdot\frac{\pi}{2}$ in the limit; the second tends to 0 by Theorem 9.3 applied to the integrable function $(g(t)-g(0+))/t$. ■

Theorem 9.6 (9.2.4, Dini's test)

Let $f \in \mathcal{R}^1[-\pi,\pi]$ and $x \in [-\pi,\pi]$. Suppose (a) $f(x+)$ and $f(x-)$ exist, and (b) $\int_0^{\delta}\frac{|f(x+t)+f(x-t)-f(x+)-f(x-)|}{t}\,dt < \infty$. Then $s_n(f;x) \to \frac{1}{2}[f(x+)+f(x-)]$.

Proof. Apply Theorem 9.4 and Lemma 9.5 to $g(t) = \frac{1}{2}[f(x+t)+f(x-t)]$, with $\lambda = n+\frac{1}{2}$. ■

Corollary 9.7 (9.2.5)

If f satisfies a Lipschitz condition at x, $|f(x+t)-f(x)| \le Mt$ for small $|t|$, then $s_n(f;x) \to f(x)$.

Proof. The integrand in (b) is at most $2M$. ■

Corollary 9.8 (9.2.6)

If f is differentiable at x, then $s_n(f;x) \to f(x)$.

Proof. Differentiability at x implies $|f(x+t)-f(x)| \le (|f'(x)|+1)|t|$ for small t. ■

Lemma 9.9 (9.2.7)

If $g:[0,\delta]\to\mathbb{R}$ is of bounded variation, then $\lim_{\lambda\to\infty}\frac{2}{\pi}\int_0^\delta g(t)\frac{\sin\lambda t}{t}\,dt = g(0+)$.

Proof. Take g increasing (Jordan) with $g(0+)=0$, and $h\in(0,\delta)$ with $g(h)<\epsilon$. On $[h,\delta]$, $g(t)/t$ is integrable, so that part $\to 0$ (Theorem 9.3). On $(0,h]$, the second mean value theorem (Corollary 5.25 of Volume I, on $[\eta,h]$, $\eta\to 0$) gives $\int_0^h g(t)\frac{\sin\lambda t}{t}dt = g(h-)\int_c^h \frac{\sin\lambda t}{t}dt$, $c\in[0,h]$; as $|\int_a^b \frac{\sin u}{u}du|\le K$ for $0\le a<b$, this is $\le \epsilon K$. ■

Theorem 9.10 (9.2.8, Dirichlet–Jordan)

If $f\in\mathcal{R}^1[-\pi,\pi]$ is of bounded variation on some interval $[x-\delta,x+\delta]$, then $s_n(f;x)\to\frac{1}{2}[f(x+)+f(x-)]$.

Proof. $g(t)=\frac{1}{2}[f(x+t)+f(x-t)]$ is of bounded variation on $[0,\delta]$ with $g(0+)=\frac{1}{2}[f(x+)+f(x-)]$; apply Theorem 9.4 and Lemma 9.9. ■

Example 9.2

For $f(x)=x$ on $(-\pi,\pi)$ the series converges to x inside, to 0 at $\pm\pi$; $x=\pi/2$ gives $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\cdots$. Near the jump partial sums overshoot by about 9% (Gibbs). Some continuous functions have Fourier series divergent at a point (du Bois-Reymond).

9.3 Fejér's theorem

Motivation. Averaged partial sums converge uniformly for every continuous periodic function.

The **Cesàro means** are $\sigma_n(f;x)=\frac{1}{n+1}\sum_{k=0}^n s_k(f;x)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)K_n(t)\,dt$ with the **Fejér kernel**

$$K_n(t)=\frac{1}{n+1}\sum_{k=0}^{n}D_k(t)=\frac{1}{n+1}\Big(\frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}}\Big)^2.$$

It satisfies $K_n\ge 0$, $\frac{1}{2\pi}\int K_n=1$, and $K_n(t)\le\frac{1}{(n+1)\sin^2(\delta/2)}\to 0$ uniformly on $\delta\le|t|\le\pi$. (The formula follows from $\sum_{k\le n}\sin(k+\frac{1}{2})t\cdot\sin\frac{t}{2}=\frac{1}{2}\sum(\cos kt-\cos(k+1)t)=\sin^2\frac{(n+1)t}{2}$.)

Theorem 9.11 (9.3.1, Fejér)

If $f\in\mathcal{R}^1[-\pi,\pi]$ and $f(x\pm)$ exist, then $\sigma_n(f;x)\to\frac{1}{2}[f(x+)+f(x-)]$. If f is continuous and 2π-periodic, $\sigma_n(f)\to f$ uniformly.

Proof. With $s=\frac{1}{2}[f(x+)+f(x-)]$ and K_n even, $\sigma_n(f;x)-s=\frac{1}{2\pi}\int_0^\pi[f(x+t)+f(x-t)-2s]K_n(t)\,dt$. Take δ with $|f(x+t)+f(x-t)-2s|<\epsilon$ on $(0,\delta)$: that part is $\le\epsilon\cdot\frac{1}{2\pi}\int_0^\pi K_n\le\epsilon/2$, and the part over $[\delta,\pi]$ is $\le\frac{1}{(n+1)\sin^2(\delta/2)}\cdot\frac{1}{2\pi}\int|f(x+t)+f(x-t)-2s|\to 0$. For continuous periodic f, δ is uniform in x and f bounded. ■

So continuous periodic functions are uniform limits of trigonometric polynomials (trigonometric Weierstrass) and are determined by their Fourier coefficients.

Lemma 9.12 (9.3.2)

If $s_n \to s$, then $\sigma_n = \frac{1}{n+1}\sum_{k\le n} s_k \to s$.

Proof. We may assume $s = 0$. Given ϵ, choose N with $|s_n| < \epsilon$ for $n \ge N$. For $n > N$, $|\sigma_n| \le \frac{1}{n+1}\sum_{k<N}|s_k| + \epsilon < 2\epsilon$ for large n. ■

Corollary 9.13 (9.3.3)

If the Fourier series of a continuous periodic f converges at x, its sum is $f(x)$.

Proof. By Lemma 9.12 the limit equals $\lim \sigma_n(f;x) = f(x)$ (Theorem 9.11). ■

Corollary 9.14 (9.3.4)

If $\int_{-\pi}^{\pi}|f|^2 < \infty$, then $\lim_n \|f - s_n(f)\|_2 = 0$.

Proof. Take continuous g vanishing near $\pm\pi$ with $\|f-g\|_2 < \epsilon$ (Theorem 7.14, Lemma 7.13) and $T = \sigma_m(g)$ with $\|g - T\|_{\sup} < \epsilon$ (Theorem 9.11), so $\|g - T\|_2 < \epsilon$. For $n \ge m$, $\|f - s_n(f)\|_2 \le \|f - T\|_2 < 2\epsilon$ (Theorem 9.1). ■

Corollary 9.15 (9.3.5, Parseval's identity)

If $I = [a, a+2\pi]$, $f \in \mathcal{R}^2(I)$ and $c_n = \frac{1}{2\pi}\int_I f(t)e^{-int}dt$, then $\frac{1}{2\pi}\int_I |f|^2 = \sum_{n\in\mathbb{Z}}|c_n|^2$; more generally $\frac{1}{2\pi}\int_I f\bar{g} = \sum c_n\overline{d_n}$ for $g \in \mathcal{R}^2(I)$ with coefficients d_n.

Proof. By periodicity the coefficients do not depend on a. From the proof of Theorem 9.1, $\|f\|_2^2 = \|f - s_n(f)\|_2^2 + \sum_{|k|\le n}|c_k|^2$; let $n \to \infty$ (Corollary 9.14). The polarized form follows from $4\langle f, g\rangle = \sum_{k=0}^{3} i^k \|f + i^k g\|^2$. ■

Example 9.3

Parseval for $f(x) = x$ gives $\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3} = 2\sum_{n\ge1}\frac{1}{n^2}$, again $\sum n^{-2} = \pi^2/6$.

9.4 Differentiation and integration of Fourier series

Motivation. Fourier series integrate termwise always, and differentiate termwise under smoothness.

Theorem 9.16 (9.4.1, termwise integration)

If $f \in \mathcal{R}^2(I)$ has Fourier series $\sum \hat{f}(n)e^{int}$, then for $a, x \in I$, $\int_a^x f(t)\,dt = \sum_n \hat{f}(n)\int_a^x e^{int}dt$, the series converging (uniformly in x).

Proof. By Cauchy–Schwarz, $\left|\int_a^x (f - s_n(f))\right| \le \sqrt{2\pi}\left(\int_I |f - s_n(f)|^2\right)^{1/2} \to 0$ by Corollary 9.14. ■

This holds even if the Fourier series of f diverges. For periodic C^1 f, $\widehat{f'}(n) = in\hat{f}(n)$; $\sum |n\hat{f}(n)|^2 < \infty$ and Cauchy–Schwarz give $\sum|\hat{f}(n)| < \infty$, so the series converges uniformly, to f (Corollary 9.13).

A sequence $a : \mathbb{Z} \to \mathbb{C}$ is **rapidly decreasing** if $|n|^k a(n)$ is bounded for every k.

Theorem 9.17 (9.4.2)

The Fourier coefficients of a C^∞ periodic function are rapidly decreasing. Conversely, if a is rapidly decreasing, $f(x) = \lim_n \sum_{|k|\le n} a(k)e^{ikx}$ defines a C^∞ periodic function with $\hat{f}(n) = a(n)$.

Proof. $\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n)$ (by parts), so $|n^k \hat{f}(n)| \le \frac{1}{2\pi}\int |f^{(k)}|$. Conversely all termwise derivatives $\sum (ik)^j a(k) e^{ikx}$ converge uniformly (M-test, $\sum |k|^j |a(k)| < \infty$), so f is C^∞ (Theorem 6.4), and termwise integration against e^{-inx} gives $\hat{f}(n) = a(n)$. ∎

Lemma 9.18 (9.4.3)

If $f : (a, b) \to \mathbb{R}$ is analytic and $K \subset (a, b)$ is compact, there are $C, R > 0$ with $|f^{(n)}(x)| \le Cn!R^n$ for $x \in K$, $n = 1, 2, \ldots$.

Proof. Fix $c \in K$; near c, $f(x) = \sum_k a_k (x - c)^k$ with positive radius, so there are $\rho > 0$, M with $|a_k|\rho^k \le M$. For $|x - c| \le \rho/2$, termwise differentiation (Theorem 6.12) gives

$$|f^{(n)}(x)| \le \sum_{k \ge n} \frac{k!}{(k-n)!} |a_k| \Big(\frac{\rho}{2}\Big)^{k-n} \le M\rho^{-n} n! \sum_{j \ge 0} \binom{n+j}{n} 2^{-j} = 2M\, n! \Big(\frac{2}{\rho}\Big)^n.$$

Finitely many such neighbourhoods cover K; take the largest constants. ∎

Lemma 9.19 (9.4.4)

If $c \ge 0$, $R > 0$, $n \ge 1$ and $c \le k!\,(R/n)^k$ for all $k = 0, 1, 2, \ldots$, then $c \le (R + 1)e^{-n/(R+1)}$.

Proof. If $c > 0$, then $\frac{1}{k!}\big(\frac{n}{R+1}\big)^k \le \frac{1}{c}\big(\frac{R}{R+1}\big)^k$ for every k; summing, $e^{n/(R+1)} \le \frac{R+1}{c}$. ∎

Theorem 9.20 (9.4.5)

If $f : \mathbb{R} \to \mathbb{C}$ is periodic and analytic, then $\limsup_{|n|\to\infty} |\hat{f}(n)|^{1/|n|} < 1$.

Proof. By Lemma 9.18 on $K = [-\pi, \pi]$, $\|f^{(k)}\|_{\sup} \le Ck!R^k$ for $k \ge 1$ (and enlarging C, also for $k = 0$). Then $|\hat{f}(n)| = |\widehat{f^{(k)}}(n)|/|n|^k \le Ck!(R/|n|)^k$ for all k, so by Lemma 9.19, $|\hat{f}(n)| \le C(R + 1)e^{-|n|/(R+1)}$, and $\limsup |\hat{f}(n)|^{1/|n|} \le e^{-1/(R+1)} < 1$. ∎

Conversely, such geometric decay of the coefficients makes f analytic (the series extends to a holomorphic function near the unit circle).

Theorem 9.21 (9.4.6)

If $\int_{-\pi}^{\pi} |f| < \infty$, then $\lim_{|n|\to\infty} \hat{f}(n) = 0$.

Proof. Theorem 9.3 applied to the real and imaginary parts of $\int f(t)(\cos nt - i \sin nt)\, dt$. ∎

The decay can be arbitrarily slow, but not every sequence tending to 0 is a sequence of Fourier coefficients:

Proposition 9.22 (9.4.7)

If $f \in \mathcal{R}^1[-\pi,\pi]$ is odd, then the partial sums of $\sum_{n\ge1}\hat f(n)/n$ are bounded.

Proof. For odd f, $\sum_{n\le N}\frac{\hat f(n)}{n} = -\frac{i}{\pi}\int_0^\pi f(t)\sum_{n\le N}\frac{\sin nt}{n}\,dt$. On $(0,\pi]$, $|\sum_{n\le N}\frac{\sin nt}{n}| \le 1+2\pi$: terms with $n \le 1/t$ are $\le t$ each, and summation by parts with (‡) of Chapter 6 bounds the rest by $2\cdot t\cdot\pi/t$. So the partial sums are $\le \frac{1+2\pi}{\pi}\int_0^\pi|f|$. ∎

So $\sum_{n\ge2}\frac{\sin nx}{\log n}$, which converges everywhere (Theorem 6.20), is not the Fourier series of any $f\in\mathcal{R}^1$, since $\sum\frac{1}{n\log n}=\infty$.

9.5 Selected exercises

Motivation. Book exercises: Lebesgue constants, regular averaging, and $1/|n|$ decay under bounded variation.

Exercise 9.5.3 (Lebesgue constants)

Show that $\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n(t)|\,dt\to\infty$ as $n\to\infty$; in fact it is at least $\frac{4}{\pi^2}(1+\frac12+\cdots+\frac1n)$.

Solution.

Step 1. D_n is even and $0<\sin\frac{t}{2}\le\frac{t}{2}$ on $(0,\pi]$, so

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n| = \frac{1}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{\sin\frac{t}{2}}\,dt \ge \frac{2}{\pi}\int_0^\pi\frac{|\sin(n+\frac12)t|}{t}\,dt = \frac{2}{\pi}\int_0^{(n+\frac12)\pi}\frac{|\sin u|}{u}\,du,$$

by the substitution $u=(n+\frac12)t$.

Step 2. On $[k\pi,(k+1)\pi]$, $\frac1u\ge\frac{1}{(k+1)\pi}$ and $\int_{k\pi}^{(k+1)\pi}|\sin u|\,du=2$. Summing over $k=0,\dots,n-1$,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}|D_n| \ge \frac{2}{\pi}\sum_{k=0}^{n-1}\frac{2}{(k+1)\pi} = \frac{4}{\pi^2}\sum_{k=1}^{n}\frac1k\to\infty.$$

By contrast $\frac{1}{2\pi}\int|K_n|=1$ ($K_n\ge0$): hence Fejér's theorem holds for all continuous f while partial sums may diverge (Example 9.2). ∎

Exercise 9.5.8 (Regular averaging methods)

Let $a_{mn}\ge0$ $(m,n\ge1)$ with $a_{mn}=0$ for $n>m$ and $\sum_{n=1}^m a_{mn}=1$ for every m. For a sequence (s_n) put $t_m=\sum_{n=1}^m a_{mn}s_n$. Show that the following are equivalent:
(a) for every convergent sequence, $\lim_n s_n=s$ implies $\lim_m t_m=s$;
(b) $\lim_{m\to\infty}a_{mn}=0$ for each n.

Solution.

(a)⇒(b): $s_n=\delta_{nn_0}\to0$ gives $t_m=a_{mn_0}\to0$.

(b)⇒(a):

Step 1. $t_m-s=\sum_{n\le m}a_{mn}(s_n-s)$, as $\sum_n a_{mn}=1$.

Step 2. If $|s_n-s|<\epsilon$ for $n>N$, then for $m>N$ $(a_{mn}\ge0)$

$$|t_m-s|\le\sum_{n\le N}a_{mn}|s_n-s|+\epsilon\sum_{N<n\le m}a_{mn}\le\sum_{n\le N}a_{mn}|s_n-s|+\epsilon.$$

Step 3. The first sum has N fixed terms, each $\to 0$ by (b); so $\limsup_m |t_m - s| \le \epsilon$.

Taking $a_{mn} = \frac{1}{m}$ for $n \le m$ recovers Lemma 9.12. ■

Exercise 9.5.13 (Fourier coefficients of functions of bounded variation)

A function $f : [-\pi, \pi] \to \mathbb{C}$ is of bounded variation if $\sup_P \sum_i |f(x_i) - f(x_{i-1})| < \infty$ over all partitions P.
(a) Show that a function of bounded variation is a linear combination of increasing real functions.
(b) Show that for every increasing $f : [-\pi, \pi] \to \mathbb{R}$, $|\hat{f}(n)| \le \frac{V_{-\pi}^{\pi}(f)}{\pi |n|}$ for $n = \pm 1, \pm 2, \dots$.
(c) Show that the Fourier coefficients of a function f of bounded variation satisfy $|\hat{f}(n)| \le \frac{C}{|n|}$ $(n \neq 0)$ for some constant $C > 0$.

Solution.

(a) $\operatorname{Re} f, \operatorname{Im} f$ are of bounded variation ($|\operatorname{Re} z|, |\operatorname{Im} z| \le |z|$); Jordan (Theorem 5.16 of Volume I) gives $\operatorname{Re} f = g_1 - h_1$, $\operatorname{Im} f = g_2 - h_2$ with g_j, h_j increasing.

(b) Let $V = V_{-\pi}^{\pi}(f) = f(\pi) - f(-\pi)$ and $n \neq 0$.

Step 1. Choose $\theta \in \mathbb{R}$ with $e^{i\theta} \int_{-\pi}^{\pi} f(t) e^{-int} dt \ge 0$. Then

$$2\pi |\hat{f}(n)| = \operatorname{Re}\Big(e^{i\theta} \int_{-\pi}^{\pi} f(t) e^{-int} dt \Big) = \int_{-\pi}^{\pi} f(t) \cos(nt - \theta)\, dt.$$

Step 2. By the second mean value theorem (Corollary 5.25 of Volume I) there is $c \in [-\pi, \pi]$ with

$$\int_{-\pi}^{\pi} f(t) \cos(nt - \theta)\, dt = f(-\pi) \int_{-\pi}^{c} \cos(nt - \theta)\, dt + f(\pi) \int_{c}^{\pi} \cos(nt - \theta)\, dt.$$

Step 3. As $\int_{-\pi}^{\pi} \cos(nt - \theta)\, dt = 0$, the right side is $-V \int_{-\pi}^{c} \cos(nt - \theta)\, dt = -\frac{V}{n} (\sin(nc - \theta) - \sin(-n\pi - \theta))$, of absolute value $\le \frac{2V}{|n|}$.

Step 4. So $2\pi |\hat{f}(n)| \le \frac{2V}{|n|}$.

(c) With $f = g_1 - h_1 + i g_2 - i h_2$ from (a), linearity and (b) give

$$|\hat{f}(n)| \le |\hat{g}_1(n)| + |\hat{h}_1(n)| + |\hat{g}_2(n)| + |\hat{h}_2(n)| \le \frac{C}{|n|},$$

with $C = \frac{1}{\pi}(V(g_1) + V(h_1) + V(g_2) + V(h_2))$, $V = V_{-\pi}^{\pi}$. $f(x) = x$ ($|\hat{f}(n)| = \frac{1}{|n|}$, Example 9.1) shows the order is sharp. ■

Chapter 10. The Lebesgue Integral

10.1 Measurable sets and Lebesgue measure

Motivation. Partitioning the range requires measuring sets $\{f > c\}$; Lebesgue measure extends length to them.

Proposition 10.1 (10.1.1)

Every open $U \subset \mathbb{R}$ is a union of countably many disjoint open intervals, and this representation is unique.

Proof. For $x \in U$, $I_x = (a_x, b_x)$ with $a_x = \inf\{a : (a, x] \subset U\}$, $b_x = \sup\{b : [x, b) \subset U\}$ is the largest open interval in U containing x. If $I_x \cap I_y \neq \emptyset$, $I_x \cup I_y \subset U$ is an open interval, so $I_x = I_y$. Distinct I_x are disjoint and contain rationals, hence countably many; they are the components of U (uniqueness). ■

For $U = \bigsqcup(a_n, b_n)$ put $\lambda(U) = \sum(b_n - a_n)$ (independent of the order, Proposition 2.16 of Volume I), and for any $S \subset \mathbb{R}$ define the **outer measure**

$$\mu(S) = \inf\{\lambda(U) : U \supset S \text{ open}\} \in [0, \infty].$$

Proposition 10.2 (10.1.2)

(a) $S \subset T \Rightarrow \mu(S) \leq \mu(T)$.
(b) If I is an interval with end points $a \leq b$, then $\mu(I) = b - a$.
(c) $\mu(x + S) = \mu(S)$.
(d) $\mu(\bigcup S_n) \leq \sum \mu(S_n)$ for countably many S_n.

Proof. First, if open intervals J_k cover (a, b), then $b - a \leq \sum |J_k|$: for $a < c < d < b$ finitely many cover $[c, d]$, a chain of overlapping ones from c to d gives $d - c \leq \sum |J_k|$; let $c \to a$, $d \to b$. Summing over components (Proposition 6.9), $\lambda(U) \leq \sum \lambda(U_n)$ for open $U \subset \bigcup U_n$.

(a) is immediate.

(b) For $I = [a, b]$, $\mu(I) \leq \lambda((a - \epsilon, b + \epsilon)) = b - a + 2\epsilon$; and an open $U \supset I$ has a component containing $[a, b]$, so $\lambda(U) \geq b - a$. Other intervals are similar.

(c) Translation preserves λ of open sets.

(d) Assume all $\mu(S_n) < \infty$; take open $U_n \supset S_n$, $\lambda(U_n) < \mu(S_n) + \epsilon 2^{-n}$: $\mu(\bigcup S_n) \leq \lambda(\bigcup U_n) \leq \sum \lambda(U_n) < \sum \mu(S_n) + \epsilon$. ■

In particular countable sets have outer measure 0, so $[0, 1]$ is uncountable. A set $E \subset \mathbb{R}$ is **measurable** if

$$\mu(X) = \mu(X \cap E) + \mu(X \cap E^c) \qquad \text{for every } X \subset \mathbb{R};$$

let $\mathfrak{M}$ be the family of measurable sets. By (d), only $\geq$ needs checking, and only for $\mu(X) < \infty$.

Theorem 10.3 (10.1.3)

(a) $\emptyset \in \mathfrak{M}$.
(b) $E \in \mathfrak{M} \Rightarrow E^c \in \mathfrak{M}$.
(c) $E, F \in \mathfrak{M} \Rightarrow E \cup F \in \mathfrak{M}$.
(d) $E_n \in \mathfrak{M} \Rightarrow \bigcup E_n \in \mathfrak{M}$.

(e) If $\mu(E) = 0$, then $E \in \mathfrak{M}$.
(f) Every interval is measurable.
(g) Every open set is measurable.

Proof. (a), (b) are clear from the symmetric definition.

(c) Since $X \cap (E \cup F) = (X \cap E) \cup (X \cap F \cap E^c)$,

$$\begin{aligned}\mu(X \cap (E \cup F)) + \mu(X \cap (E \cup F)^c) &\le \mu(X \cap E) + \mu(X \cap E^c \cap F) + \mu(X \cap E^c \cap F^c) \\ &= \mu(X \cap E) + \mu(X \cap E^c) = \mu(X),\end{aligned}$$

using $F \in \mathfrak{M}$ (for the set $X \cap E^c$) and then $E \in \mathfrak{M}$.

(d) $F_n = E_n - (E_1 \cup \cdots \cup E_{n-1})$ are measurable ((b), (c), $A \cap B = (A^c \cup B^c)^c$) and disjoint; $G_n = F_1 \cup \cdots \cup F_n$, $E = \bigsqcup F_n$. Testing F_n on $X \cap G_n$, $\mu(X \cap G_n) = \mu(X \cap G_{n-1}) + \mu(X \cap F_n) = \sum_{i \le n} \mu(X \cap F_i)$. As $G_n \in \mathfrak{M}$, $G_n^c \supset E^c$,

$$\mu(X) = \mu(X \cap G_n) + \mu(X \cap G_n^c) \ge \sum_{i \le n} \mu(X \cap F_i) + \mu(X \cap E^c).$$

Let $n \to \infty$ and use (d) of Proposition 10.2: $\mu(X) \ge \mu(X \cap E) + \mu(X \cap E^c)$.

(e) $\mu(X \cap E) = 0$ and $\mu(X \cap E^c) \le \mu(X)$.

(f) By (b), (d), (e), take $I = (a, \infty)$ and open $U \supset X$ with $\lambda(U) < \mu(X) + \epsilon$. Splitting components, $\lambda(U \cap (a,\infty)) + \lambda(U \cap (-\infty, a)) \le \lambda(U)$, so $\mu(X \cap I) + \mu(X \cap I^c) \le \lambda(U \cap (a,\infty)) + \lambda(U \cap (-\infty, a)) + \mu(\{a\}) < \mu(X) + \epsilon$.

(g) Proposition 10.1, (f) and (d). ■

So $\mathfrak{M}$ is a σ-algebra containing all Borel sets. The **Lebesgue measure** is $m = \mu|_{\mathfrak{M}}$.

Theorem 10.4 (10.1.4, countable additivity)

If $E_n \in \mathfrak{M}$ are disjoint, then $m(\bigsqcup E_n) = \sum m(E_n)$.

Proof. In the proof of Theorem 10.3(d) take $X = E = \bigsqcup E_n$ (so $F_n = E_n$): $m(E) \ge \sum m(E_n)$; the reverse is Proposition 10.2(d). ■

Hence $m(\bigcup E_n) = \lim m(E_n)$ for $E_n \uparrow$ (use $\bigsqcup(E_n - E_{n-1})$), and $m(\bigcap E_n) = \lim m(E_n)$ for $E_n \downarrow$ with $m(E_1) < \infty$. Also $m(E) = \inf\{m(U) : U \supset E \text{ open}\}$, and for bounded E, $m(E) = \sup\{m(K) : K \subset E \text{ compact}\}$ (outer approximation of $[a,b] - E$). Not every set is measurable (Vitali): one point of $[0,1)$ per coset of $\mathbb{Q}$ gives V whose rational translates mod 1 partition $[0,1)$, so $m(V)$ can be neither 0 nor positive.

10.2 Measurable functions and the integral

Motivation. Integrate simple functions, then approximate; measurable functions are closed under limits.

Let $E \in \mathfrak{M}$ and $\mathbb{R}^* = [-\infty, \infty]$. $f : E \to \mathbb{R}^*$ is **measurable** if $\{x \in E : f(x) > a\} \in \mathfrak{M}$ for every $a \in \mathbb{R}$; a complex function is measurable if its real and imaginary parts are.

Proposition 10.5 (10.2.1)

For $f : E \to \mathbb{R}^*$ the following are equivalent:
(a) f is measurable;
(b) $\{f \geq a\} \in \mathfrak{M}$ for all a;
(c) $\{f < a\} \in \mathfrak{M}$ for all a;
(d) $\{f \leq a\} \in \mathfrak{M}$ for all a.

Proof. $\{f \geq a\} = \bigcap_n \{f > a - 1/n\}$, $\{f > a\} = \bigcup_n \{f \geq a + 1/n\}$, and (c), (d) are the complements (relative to E) of (b), (a). ∎

Corollary 10.6 (10.2.2)

If f is measurable, then $\{f = a\}$ is measurable for every $a \in \mathbb{R}^*$.

Proof. $\{f = a\} = \{f \geq a\} \cap \{f \leq a\}$ for real a, and $\{f = \infty\} = \bigcap_n \{f \geq n\}$. ∎

Proposition 10.7 (10.2.3)

(a) Continuous $f : E \to \mathbb{R}$ are measurable.
(b) If f, g are measurable and real-valued and $a \in \mathbb{R}$, then $f + g$, fg, af are measurable.
(c) If each f_n is measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf f_n$, $\limsup f_n$, and hence $\lim f_n$ where it exists.

Proof. (a) $\{f > a\} = U \cap E$ with U open.

(b) $\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} \{f > r\} \cap \{g > a - r\}$; f^2 is measurable since $\{f^2 > a\} = \{f > \sqrt{a}\} \cup \{f < -\sqrt{a}\}$ for $a \geq 0$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

(c) $\{\sup f_n > a\} = \bigcup \{f_n > a\}$, $\inf f_n = -\sup(-f_n)$, $\limsup f_n = \inf_n \sup_{k \geq n} f_k$. ∎

A **simple function** is $s = \sum_{i=1}^k a_i \chi_{A_i}$ with $a_i \geq 0$ and $A_i \in \mathfrak{M}$; its integral is $\int_E s = \sum a_i m(A_i \cap E)$ (independent of the representation, by passing to a common disjoint refinement and using additivity). For measurable $f : E \to [0, \infty]$,

$$\int_E f = \sup\Big\{ \int_E s : 0 \leq s \leq f,\ s \text{ simple} \Big\}.$$

For simple functions the two definitions agree, and $0 \leq f \leq g$ implies $\int f \leq \int g$.

Proposition 10.8 (10.2.4)

Every measurable $f : E \to [0, \infty]$ is the pointwise limit of an increasing sequence of simple functions.

Proof. Let $A_{n,i} = \{\frac{i-1}{2^n} \leq f < \frac{i}{2^n}\}$ $(1 \leq i \leq n2^n)$, $B_n = \{f \geq n\}$, and $s_n = \sum_i \frac{i-1}{2^n} \chi_{A_{n,i}} + n\chi_{B_n}$. Then $s_n \leq s_{n+1} \leq f$, and $f - s_n < 2^{-n}$ where $f < n$; also $s_n = n \to \infty$ where $f = \infty$. ∎

Theorem 10.9 (10.2.5)

If $A = \bigsqcup A_n$ with $A_n \in \mathfrak{M}$ disjoint and $f : A \to [0, \infty]$ is measurable, then $\int_A f = \sum_n \int_{A_n} f$.

Proof. For simple $s \leq f$, $\int_A s = \sum_n \int_{A_n} s \leq \sum \int_{A_n} f$ (Theorem 10.4); take the supremum. Conversely, simple $s_k \leq f$ on A_k with $\int_{A_k} s_k > \int_{A_k} f - \epsilon/N$ $(k \leq N)$ give $s = \sum_{k \leq N} s_k \chi_{A_k} \leq f$, so $\int_A f \geq \sum_{k \leq N} \int_{A_k} f - \epsilon$. ∎

So $A \mapsto \int_A f$ is a measure. For measurable f, write $f^+ = \max(f, 0)$, $f^- = \max(-f, 0)$; $f \in L^1(E)$ if $\int_E |f| < \infty$, and then $\int_E f = \int_E f^+ - \int_E f^-$; for complex f, $\int f = \int \operatorname{Re} f + i \int \operatorname{Im} f$.

Corollary 10.10 (10.2.6)

Theorem 10.9 holds for every $f \in L^1(A)$, the series converging absolutely.

Proof. Apply Theorem 10.9 to $f^{\pm}$ (real and imaginary parts) and subtract; the series converge because $\sum \int_{A_n} |f| = \int_A |f| < \infty$. ■

Corollary 10.11 (10.2.7)

A measurable f is in $L^1(E)$ iff $|f| \in L^1(E)$, and then $|\int_E f| \le \int_E |f|$.

Proof. The first statement is the definition. With θ such that $e^{-i\theta} \int f = |\int f|$, linearity (Theorem 10.14) gives

$$\left| \int f \right| = \int \operatorname{Re}(e^{-i\theta} f) \le \int |f|.$$

■

Proposition 10.12 (10.2.8)

If $f \ge 0$ is measurable and $\int_E f = 0$, then $f = 0$ almost everywhere.

Proof. With $A_n = \{f \ge 1/n\}$, $\frac{1}{n} m(A_n) \le \int f = 0$, so $m(\{f > 0\}) = m(\bigcup A_n) = 0$. ■

10.3 Convergence theorems

Motivation. Limits and Lebesgue integrals commute under mild conditions, without uniform convergence.

Theorem 10.13 (10.3.1, monotone convergence)

If $0 \le f_1 \le f_2 \le \cdots$ are measurable on E and $f = \lim f_n$, then $\int_E f = \lim \int_E f_n$.

Proof. f is measurable and $\int f_n \le \int f_{n+1} \le \int f$, so $L = \lim \int f_n \le \int f$. Let $0 \le s \le f$ be simple and $0 < c < 1$; $E_n = \{f_n \ge cs\}$ increase to E (where $f(x) = 0$, $x \in E_1$; otherwise $cs(x) < f(x)$). Then $\int f_n \ge \int_{E_n} f_n \ge c \int_{E_n} s \to c \int_E s$ (Theorem 10.9, continuity from below). So $L \ge c \int s$; let $c \to 1$ and take the supremum over s. ■

Theorem 10.14 (10.3.2, linearity)

(a) If $f, g \in L^1(E)$, then $f + g \in L^1(E)$; if $f, g \ge 0$ or $f, g \in L^1(E)$, then $\int (f + g) = \int f + \int g$.
(b) If $f \in L^1(E)$ and $a \in \mathbb{C}$, then $af \in L^1(E)$ and $\int af = a \int f$.

Proof. Clear for simple functions. For $f, g \ge 0$, simple $s_n \uparrow f$, $t_n \uparrow g$ (Proposition 10.8) give $s_n + t_n \uparrow f + g$; apply Theorem 10.13. Integrability: $|f + g| \le |f| + |g|$. For real $f, g \in L^1$, $h = f + g$: integrate $h^+ + f^- + g^- = h^- + f^+ + g^+$. Complex: real and imaginary parts.

(b) is similar (for $a = i$ the parts are exchanged). ■

Corollary 10.15 (10.3.3)

If $f_n : E \to [0,\infty]$ are measurable, then $\int_E \sum_n f_n = \sum_n \int_E f_n$.

Proof. Apply Theorem 10.13 to the partial sums, using Theorem 10.14. ■

Theorem 10.16 (10.3.4, Riemann and Lebesgue integrals)

Let $f : [a,b] \to \mathbb{R}$ be bounded.
(a) If f is Riemann integrable, then f is measurable and its Lebesgue integral equals its Riemann integral.
(b) f is Riemann integrable iff the set of its discontinuities has measure 0.

Proof. Take partitions $P_1 \subset P_2 \subset \cdots$, $\|P_k\| \to 0$, and step functions ℓ_k, u_k equal to m_i, M_i on the open subintervals of P_k (f at partition points). Then $\ell_k \uparrow \ell$, $u_k \downarrow u$, $\ell \le f \le u$, and monotone convergence (on $u_1 - u_k$, $\ell_k - \ell_1$) gives $\int \ell = \lim L(f,P_k)$, $\int u = \lim U(f,P_k)$. Off the countable $P = \bigcup P_k$, $u(x), \ell(x)$ are limits of $\sup f$, $\inf f$ over shrinking neighbourhoods, so f is continuous at x iff $u(x) = \ell(x)$.

If f is Riemann integrable, $U(f,P_k) - L(f,P_k) \to 0$ (proof of Theorem 5.8 of Volume I), so $\int (u-\ell) = 0$, $u = \ell$ a.e. (Proposition 10.12), $f = u$ a.e. is measurable (Theorem 10.3(e)), and $\int f = \int u = \lim U(f,P_k)$: this gives (a) and necessity in (b). Conversely, null discontinuity set gives $u = \ell$ a.e., so $U(f,P_k) - L(f,P_k) \to \int (u-\ell) = 0$ and Riemann's criterion applies. ■

Theorem 10.17 (10.3.5, Fatou's lemma)

If $f_n : E \to [0,\infty]$ are measurable, then $\int_E \liminf f_n \le \liminf \int_E f_n$.

Proof. $g_n = \inf_{k \ge n} f_k$ increases to $\liminf f_n$ and $\int g_n \le \int f_k$ for $k \ge n$; apply Theorem 10.13. ■

Theorem 10.18 (10.3.6, Lebesgue's dominated convergence theorem)

Let $f_n : E \to \mathbb{C}$ be measurable with (a) $|f_n| \le g$ for some $g \in L^1(E)$ and (b) $f(x) = \lim f_n(x)$ for each x. Then $f \in L^1(E)$ and $\lim \int_E |f_n - f| = 0$; in particular $\int f_n \to \int f$.

Proof. $|f| \le g$, so $f \in L^1$. Apply Fatou to $2g - |f_n - f| \ge 0$: $\int 2g \le \liminf \int (2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. Hence $\limsup \int |f_n - f| \le 0$. ■

The theorems remain valid when the hypotheses hold only almost everywhere, since sets of measure 0 do not affect integrals.

Theorem 10.19 (10.3.7, Beppo Levi)

If $f_n \in L^1(E)$ and $\sum \int_E |f_n| < \infty$, then $\sum f_n(x)$ converges absolutely for almost every x, its sum f is in $L^1(E)$, and $\int_E f = \sum \int_E f_n$.

Proof. $g = \sum |f_n|$ has $\int g = \sum \int |f_n| < \infty$ (Corollary 10.15), so $g < \infty$ a.e. There the series converges absolutely, with partial sums dominated by g; apply Theorem 10.18. ■

Theorem 10.20 (10.3.8, absolute continuity)

If $f \in L^1(E)$, then for every $\epsilon > 0$ there is $\delta > 0$ with $\int_A |f| < \epsilon$ for all measurable $A \subset E$ with $m(A) < \delta$.

Proof. $\min(|f|,n)\uparrow|f|$, so by monotone convergence choose N with $\int(|f|-\min(|f|,N))<\epsilon/2$, and $\delta=\epsilon/(2N)$. Then $\int_A|f|\le\epsilon/2+Nm(A)<\epsilon$. ∎

10.4 The spaces L^1 and L^2

Motivation. L^1 and L^2 are complete, giving the clean form of Chapter 9's Fourier theory.

Identifying functions that agree a.e., $L^1(E)$ with $\|f\|_1=\int|f|$ and $L^2(E)$ (measurable f with $\int|f|^2<\infty$) with $\|f\|_2=(\int|f|^2)^{1/2}$ are normed spaces; L^2 has the inner product $\langle f,g\rangle=\int f\bar g$ ($|f\bar g|\le\frac12(|f|^2+|g|^2)$).

Theorem 10.21 (10.4.1)

Every Cauchy sequence in $L^1(E)$ converges.

Proof. By Proposition 7.11 it suffices that $\sum\|f_n\|_1<\infty$ implies convergence of $\sum f_n$ in L^1. By Theorem 10.19, $f=\sum f_n$ exists a.e. and is in L^1, and $\|f-\sum_{n\le N}f_n\|_1\le\sum_{n>N}\|f_n\|_1\to0$. ∎

Theorem 10.22 (10.4.2)

$L^2(E)$ is a Banach space (a Hilbert space).

Proof. Let $\sum\|f_n\|_2=M<\infty$, $g=\sum|f_n|$. $\|\sum_{n\le N}|f_n|\|_2\le M$ and monotone convergence give $\int g^2\le M^2$, so $f=\sum f_n$ exists a.e., $|f|\le g$. As $|f-\sum_{n\le N}f_n|^2\le g^2$ tends to 0 a.e., dominated convergence gives L^2 convergence; apply Proposition 7.11. ∎

Theorem 10.23 (10.4.3, Riesz–Fischer)

For every $c\in\ell^2(\mathbb{Z})$ there is a periodic $f\in L^2[-\pi,\pi]$ with $\hat f(n)=c(n)$ for all n.

Proof. By orthonormality, $s_N=\sum_{|n|\le N}c(n)e^{inx}$ satisfies $\frac{1}{2\pi}\int|s_M-s_N|^2=\sum_{N<|n|\le M}|c(n)|^2$, so (s_N) is Cauchy in L^2 and converges to some f (Theorem 10.22). By Cauchy–Schwarz, $\hat f(n)=\lim_N\widehat{s_N}(n)=c(n)$. ∎

Thus $f\mapsto\hat f$ is an isometric isomorphism of $L^2[-\pi,\pi]$ (with $\frac{1}{2\pi}\int$) onto $\ell^2(\mathbb{Z})$, once injectivity is known (Theorem 10.26).

Proposition 10.24 (10.4.4)

If $f\in L^1(E)$ and $\int_A f=0$ for every measurable $A\subset E$, then $f=0$ a.e.

Proof. With $A=\{\operatorname{Re}f>0\}$, $\int_A\operatorname{Re}f=\operatorname{Re}\int_A f=0$, so $\operatorname{Re}f\chi_A=0$ a.e. (Proposition 10.12) and $m(A)=0$. Similarly for $\{\operatorname{Re}f<0\}$, $\{\operatorname{Im}f>0\}$, $\{\operatorname{Im}f<0\}$. ∎

Lemma 10.25 (10.4.5)

For every measurable $E\subset[a,b]$ there are continuous g_n with $|g_n|\le1$ and $g_n\to\chi_E$ a.e.

Proof. By regularity take compact $K_n\subset E\subset U_n$ open, $m(U_n-K_n)<2^{-n}$, and $g_n(x)=\frac{d(x,U_n^c)}{d(x,U_n^c)+d(x,K_n)}$: continuous, 1 on K_n, 0 off U_n, so $g_n\ne\chi_E$ only on U_n-K_n. Points in infinitely many U_n-K_n lie in $\bigcup_{n\ge N}(U_n-K_n)$ (measure $\le2^{1-N}$) for all N, a null set; elsewhere $g_n\to\chi_E$. ∎

Theorem 10.26 (10.4.6)

If $f \in L^1[-\pi,\pi]$ and $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$ a.e.

Proof. $\int fg = 0$ for trigonometric polynomials g, hence for continuous periodic g (Fejér, Theorem 9.11, and dominated convergence). For measurable $A \subset [-\pi+\eta, \pi-\eta]$, Lemma 10.25 (with $U_n \subset (-\pi,\pi)$) gives continuous periodic $g_n \to \chi_A$ a.e., $|g_n| \le 1$, so $\int_A f = \lim \int f g_n = 0$. Let $\eta \to 0$ and apply Proposition 10.24. ■

Hence two L^1 functions with the same Fourier coefficients agree a.e., and Parseval's identity holds for every $f \in L^2[-\pi,\pi]$.

Lemma 10.27 (10.4.7)

For $f \in L^2(E)$ and $\epsilon > 0$ there is a simple function s (a finite complex combination of characteristic functions of sets of finite measure) with $\|f - s\|_2 < \epsilon$.

Proof. For $f \ge 0$, simple $s_n \uparrow f$ (Proposition 10.8) have $|f - s_n|^2 \le f^2$, so $\|f - s_n\|_2 \to 0$ (dominated convergence), and $\int s_n^2 < \infty$ forces finite-measure sets. In general use $f = (\operatorname{Re} f)^+ - (\operatorname{Re} f)^- + i(\operatorname{Im} f)^+ - i(\operatorname{Im} f)^-$. ■

Theorem 10.28 (10.4.8)

For $f \in L^2[a,b]$ and $\epsilon > 0$ there is $g \in C[a,b]$ with $\|f - g\|_2 < \epsilon$.

Proof. By Lemma 10.27 it suffices to approximate χ_A, $A \subset [a,b]$ measurable. With g_n from Lemma 10.25, $|g_n - \chi_A|^2 \le 4$ and tends to 0 a.e., so $\|g_n - \chi_A\|_2 \to 0$ by dominated convergence on the finite interval. ■

Theorem 10.29 (10.4.9)

For $f \in L^1(E)$ and $\epsilon > 0$ there is a continuous g on $\mathbb{R}$ vanishing outside a bounded interval with $\|f - g\|_1 < \epsilon$ (norm taken over E).

Proof. By dominated convergence $\|f - f\chi_{[-N,N]}\|_1 \to 0$; approximate $f\chi_{[-N,N]}$ in L^1 by simple functions as in Lemma 10.27; approximate each $\chi_{A\cap E}$, $A \subset [-N,N]$, by g_n from Lemma 10.25 (with $U_n \subset (-N-1, N+1)$), using dominated convergence. ■

In particular L^1 and L^2 are the completions of the spaces $\mathcal{R}^1$, $\mathcal{R}^2$ of Chapter 7.

10.5 The Poisson summation formula

Motivation. Poisson summation links Fourier series with integrals over $\mathbb{R}$.

For $g \in L^1(\mathbb{R})$ let $G(x) = \sum_{n\in\mathbb{Z}} g(x + 2\pi n)$.

Theorem 10.30 (10.5.1)

The series defining G converges absolutely for almost every x, $G \in L^1[-\pi,\pi]$, and

$$\hat{G}(n) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

Proof. Integrals over $\mathbb{R}$ are translation invariant (Proposition 10.2(c), simple functions, monotone convergence). By Corollary 10.15, $\int_{-\pi}^{\pi} \sum_n |g(x+2\pi n)|\,dx = \sum_n \int_{-\pi+2\pi n}^{\pi+2\pi n} |g| = \int_{\mathbb{R}} |g| < \infty$. So the series converges absolutely a.e. and $G \in L^1$. By Theorem 10.19,

$$\hat{G}(n) = \frac{1}{2\pi} \sum_k \int_{-\pi}^{\pi} g(x+2\pi k) e^{-inx} dx = \frac{1}{2\pi} \sum_k \int_{-\pi+2\pi k}^{\pi+2\pi k} g(y) e^{-iny} dy = \frac{1}{2\pi} \int_{\mathbb{R}} g e^{-iny} dy.$$

■

If the Fourier series of G converges to $G(0)$ at 0 (e.g. g smooth, rapidly decreasing), this is the **Poisson summation formula** $\sum_n g(2\pi n) = \frac{1}{2\pi} \sum_n \int_{\mathbb{R}} g(x) e^{-inx} dx$; for $g(x) = e^{-ax^2}$ it gives $\theta(1/x) = \sqrt{x}\,\theta(x)$, $\theta(x) = \sum_n e^{-\pi n^2 x}$.